

BFS Traversal

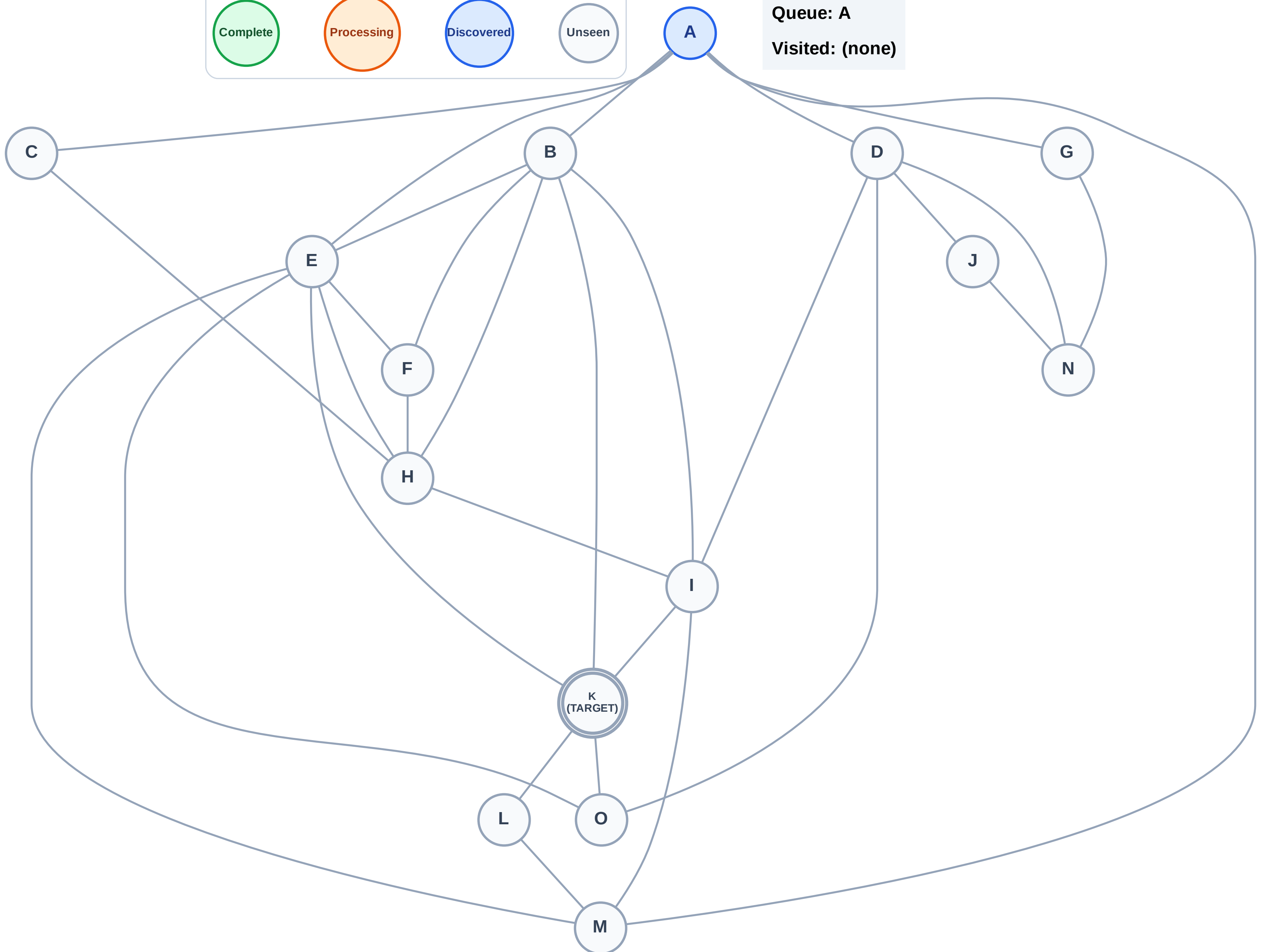
Step 1: Start at A (Seeking target node K)

Legend



Queue: A

Visited: (none)



BFS Traversal

Step 2: Process node A

Legend

Complete

Processing

Discovered

Unseen

Queue: (empty)
Visited: (none)



BFS Traversal

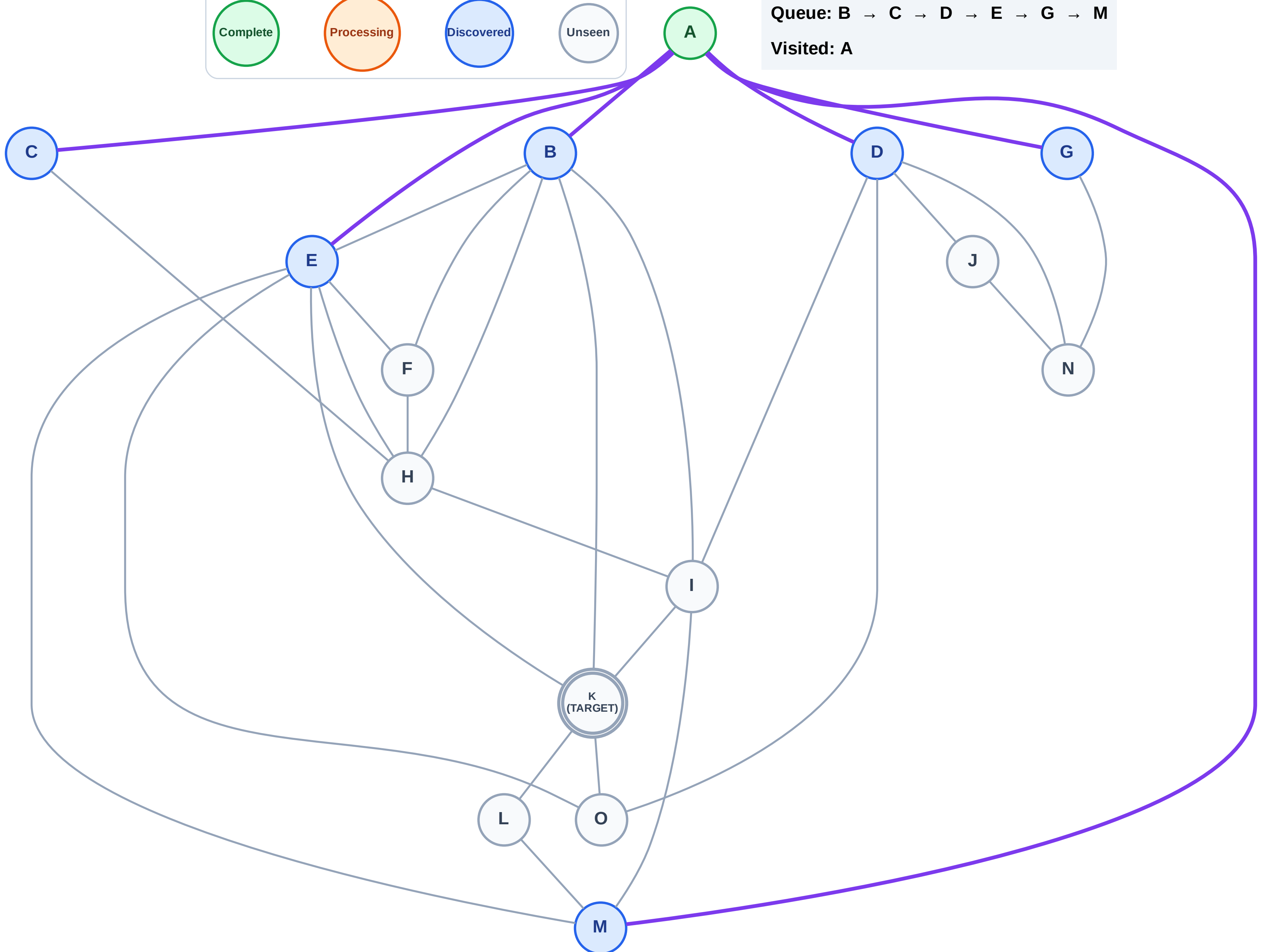
Step 3: Discover B, C, D, E, G, M from A

Legend



Queue: B → C → D → E → G → M

Visited: A



BFS Traversal

Step 4: Process node B

Legend

Complete

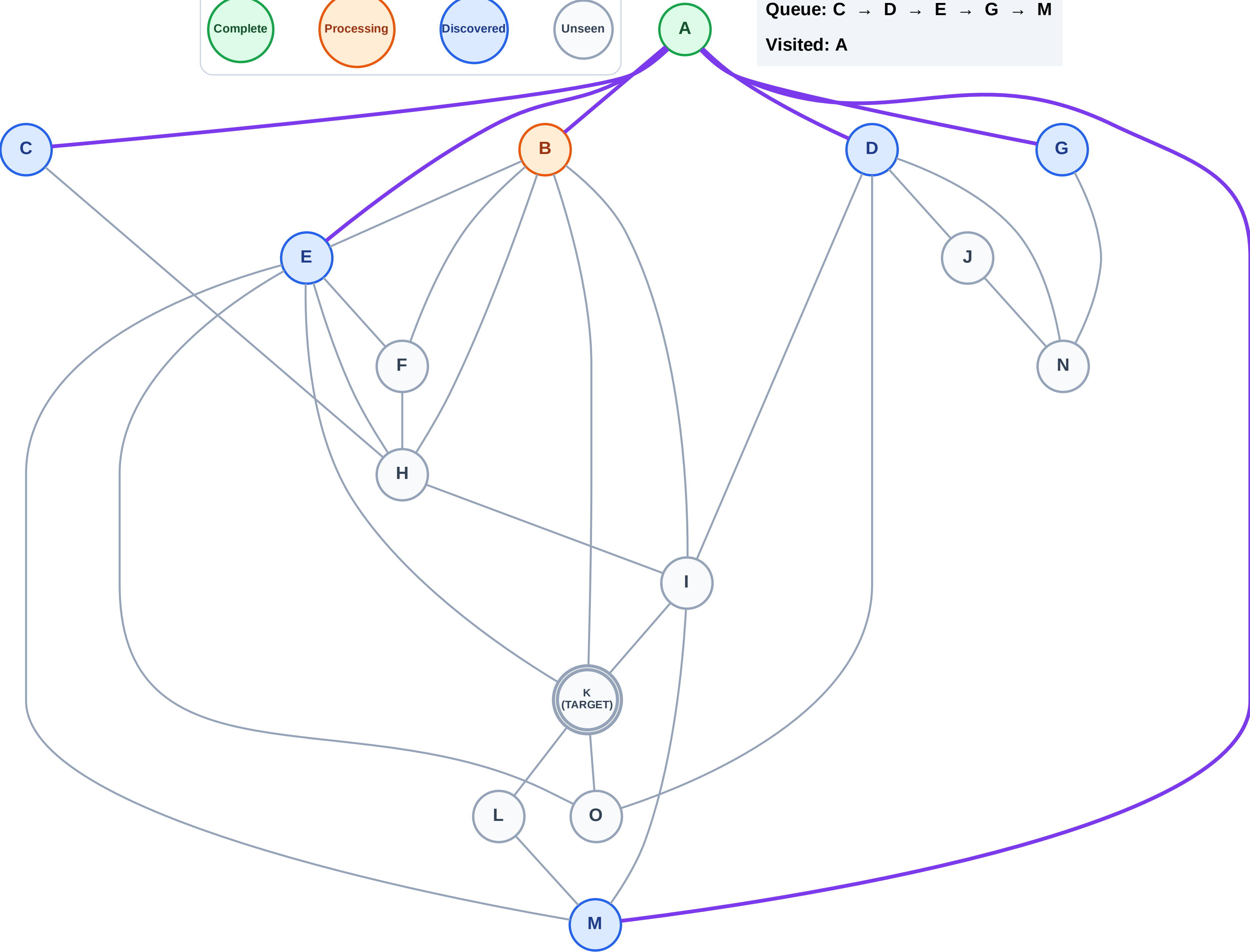
Processing

Discovered

Unseen

Queue: C → D → E → G → M

Visited: A



BFS Traversal

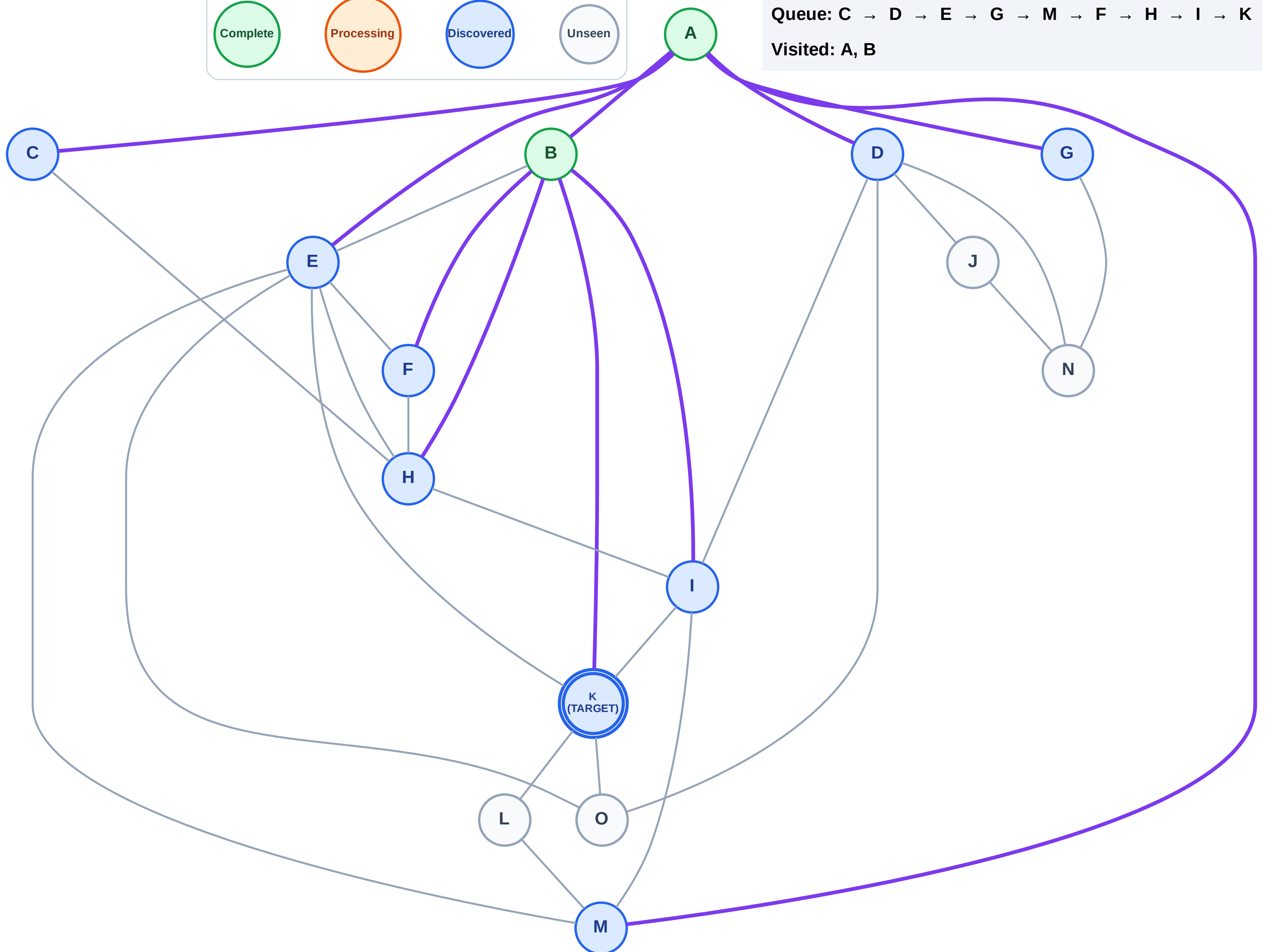
Step 5: Discover F, H, I, K from B

Legend



Queue: C → D → E → G → M → F → H → I → K

Visited: A, B



BFS Traversal

Step 6: Process node C

Legend

Complete

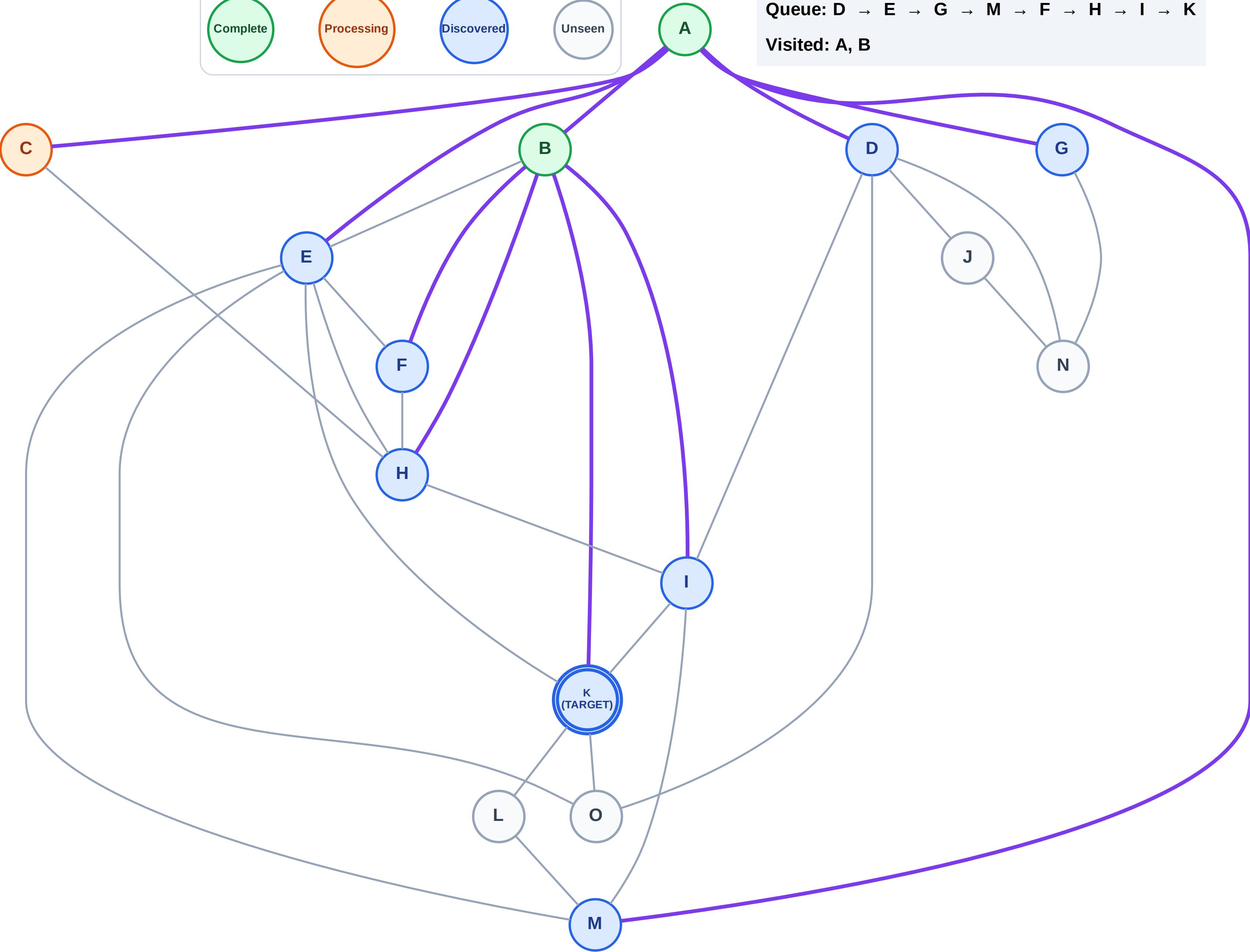
Processing

Discovered

Unseen

Queue: D → E → G → M → F → H → I → K

Visited: A, B



BFS Traversal

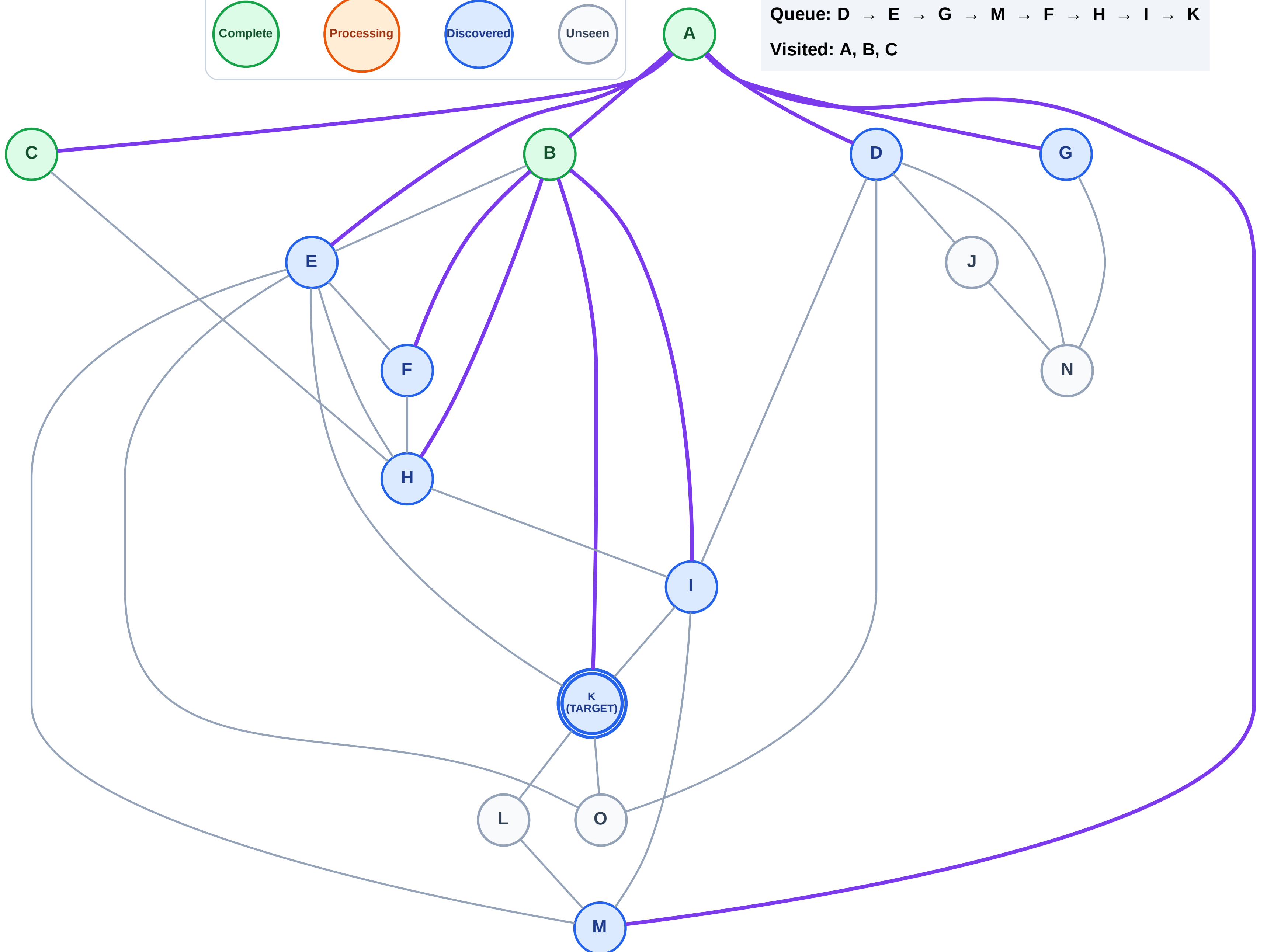
Step 7: No new neighbors from C

Legend



Queue: D → E → G → M → F → H → I → K

Visited: A, B, C



BFS Traversal

Step 8: Process node D

Legend

Complete

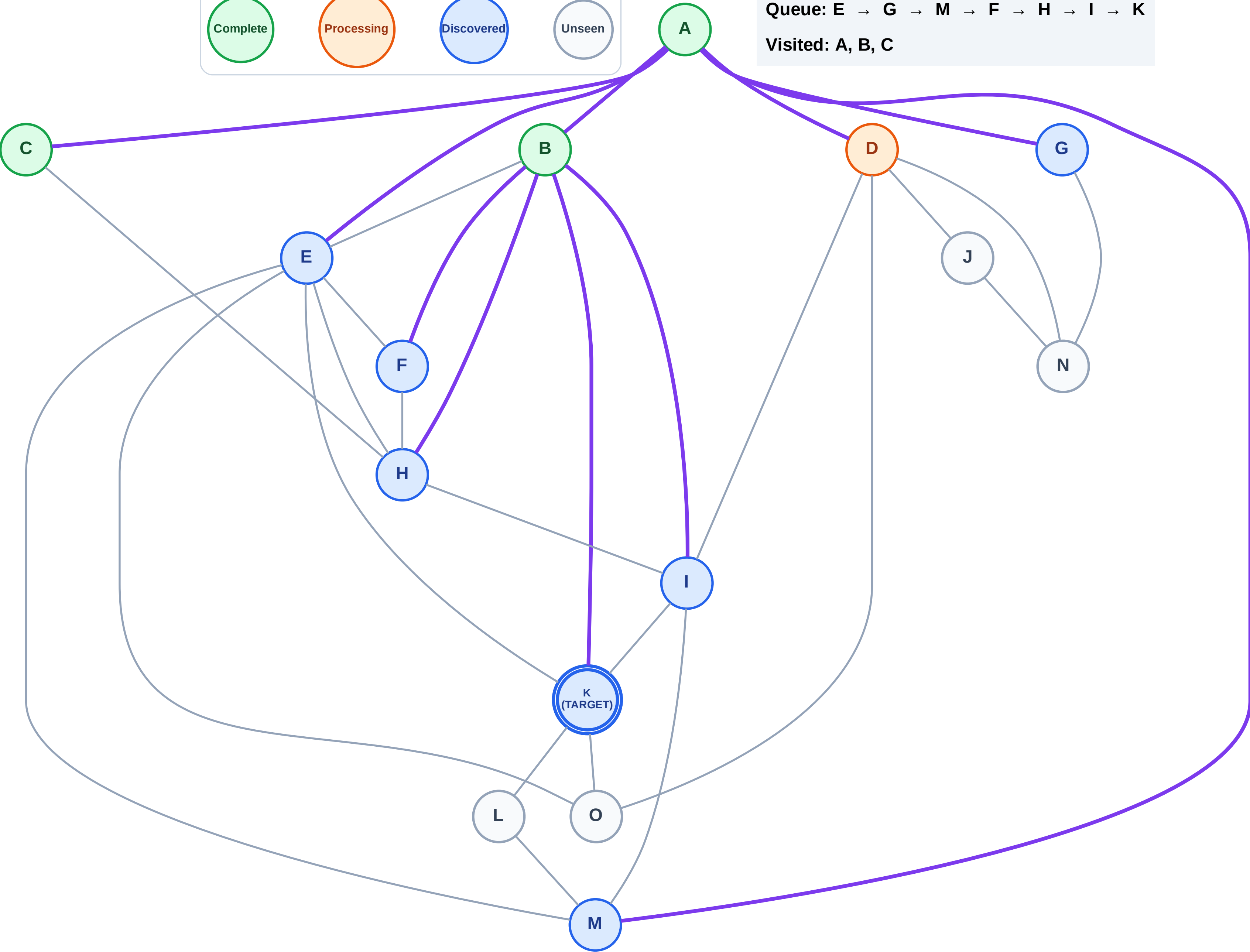
Processing

Discovered

Unseen

Queue: E → G → M → F → H → I → K

Visited: A, B, C



BFS Traversal

Step 9: Discover J, N, O from D

Legend



Complete



Processing



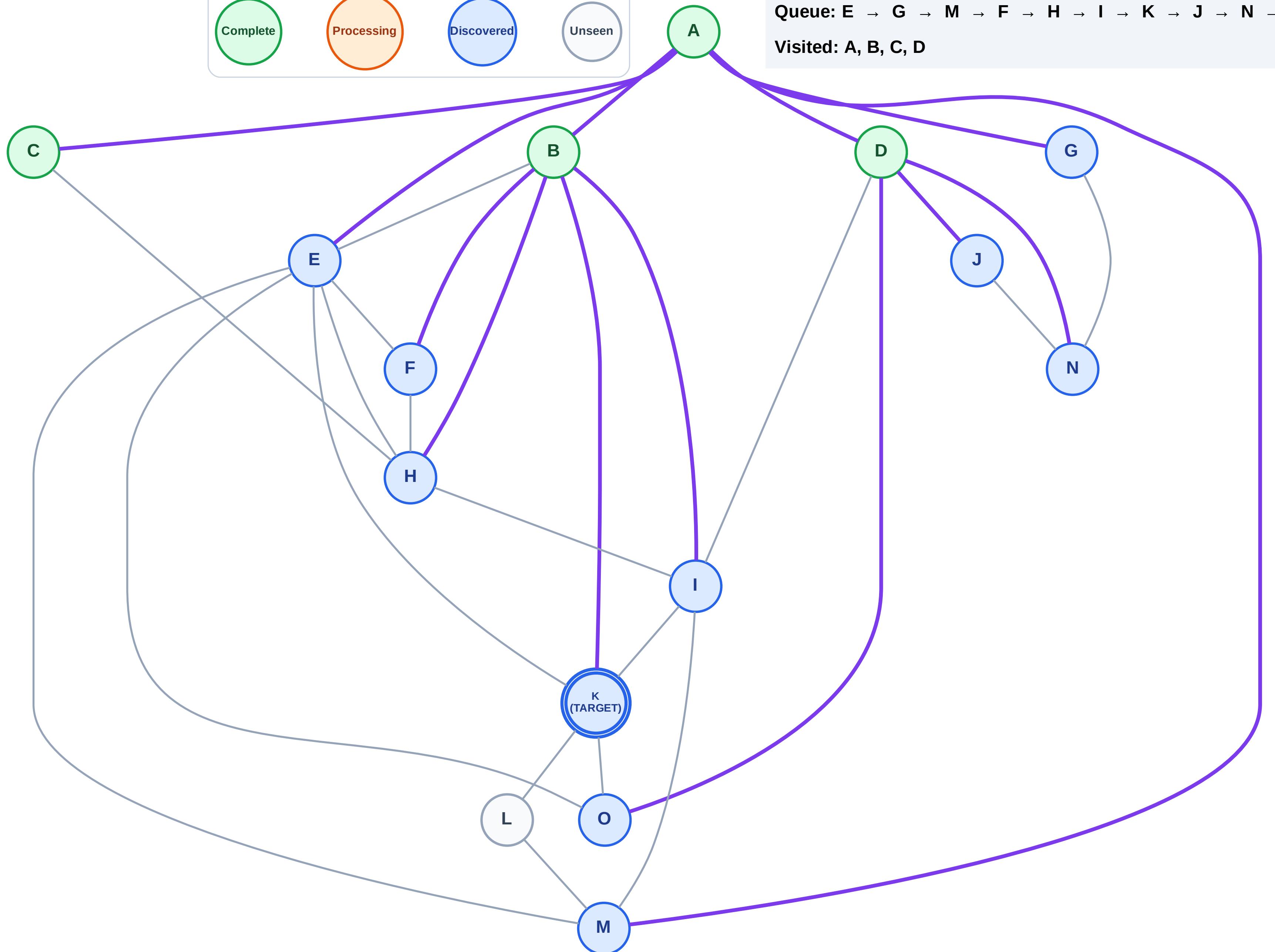
Discovered



Unseen

Queue: E → G → M → F → H → I → K → J → N → O

Visited: A, B, C, D



BFS Traversal

Step 10: Process node E

Legend



Complete



Processing



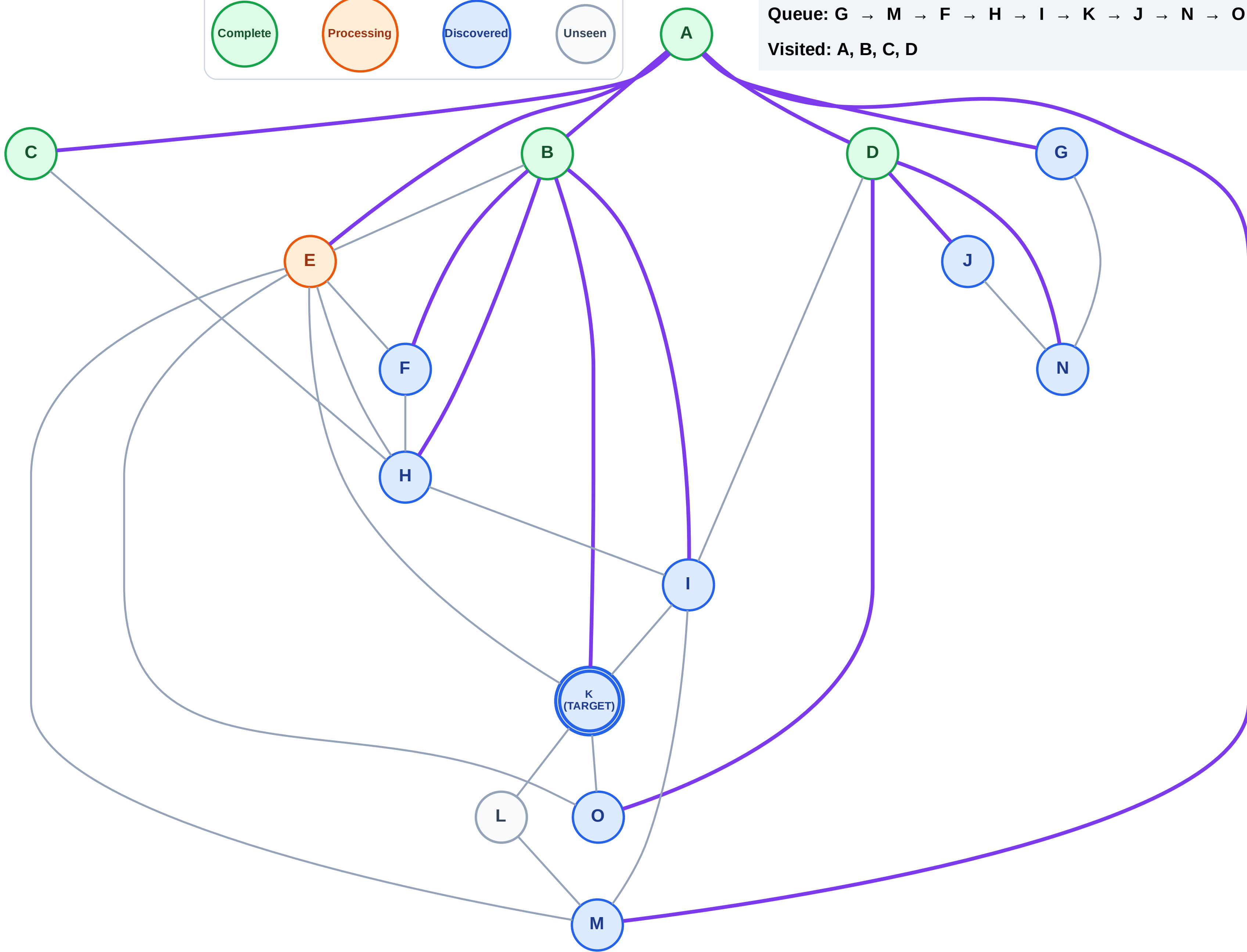
Discovered



Unseen

Queue: G → M → F → H → I → K → J → N → O

Visited: A, B, C, D



BFS Traversal

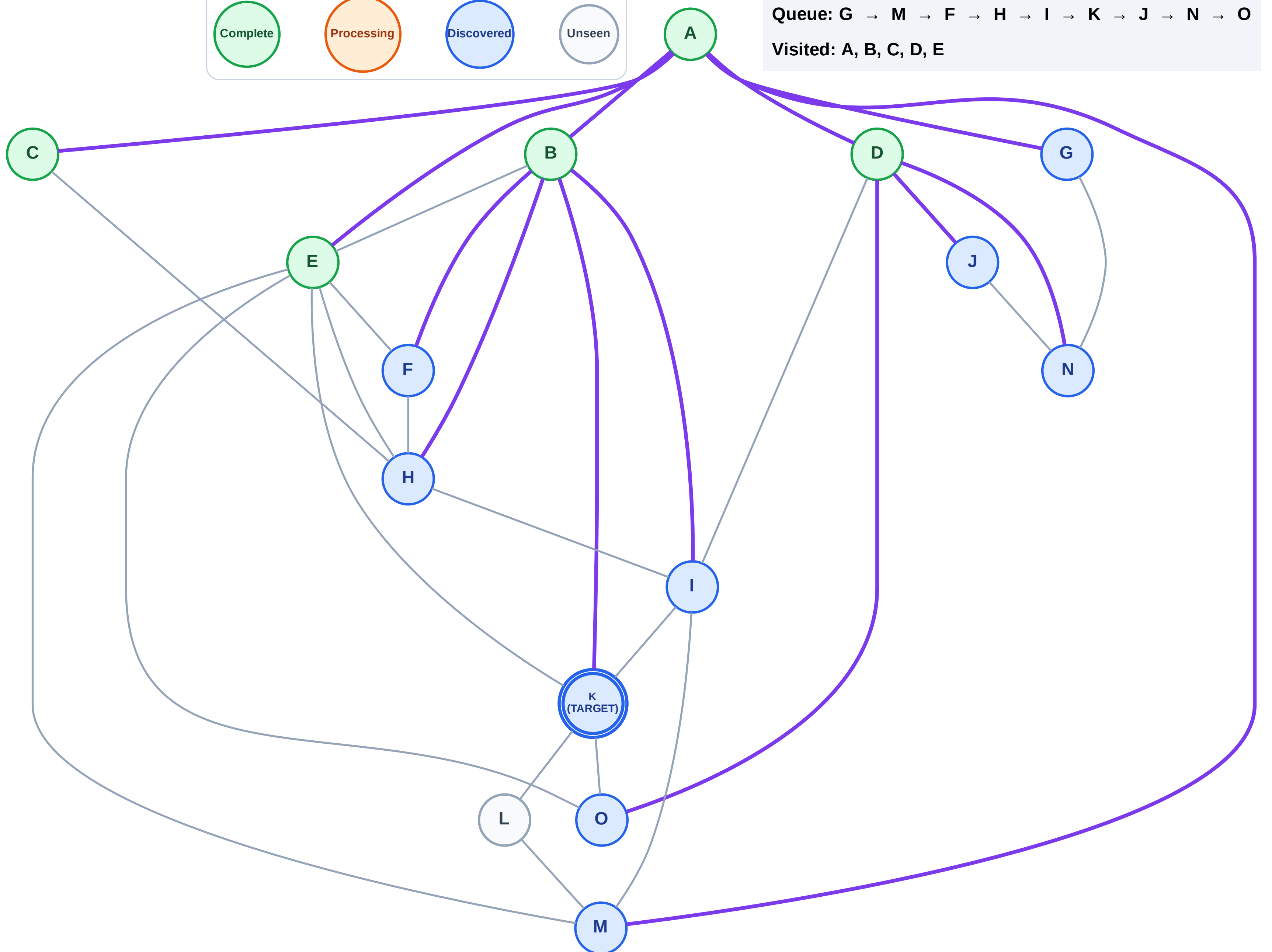
Step 11: No new neighbors from E

Legend



Queue: G → M → F → H → I → K → J → N → O

Visited: A, B, C, D, E



BFS Traversal

Step 12: Process node G

Legend

Complete

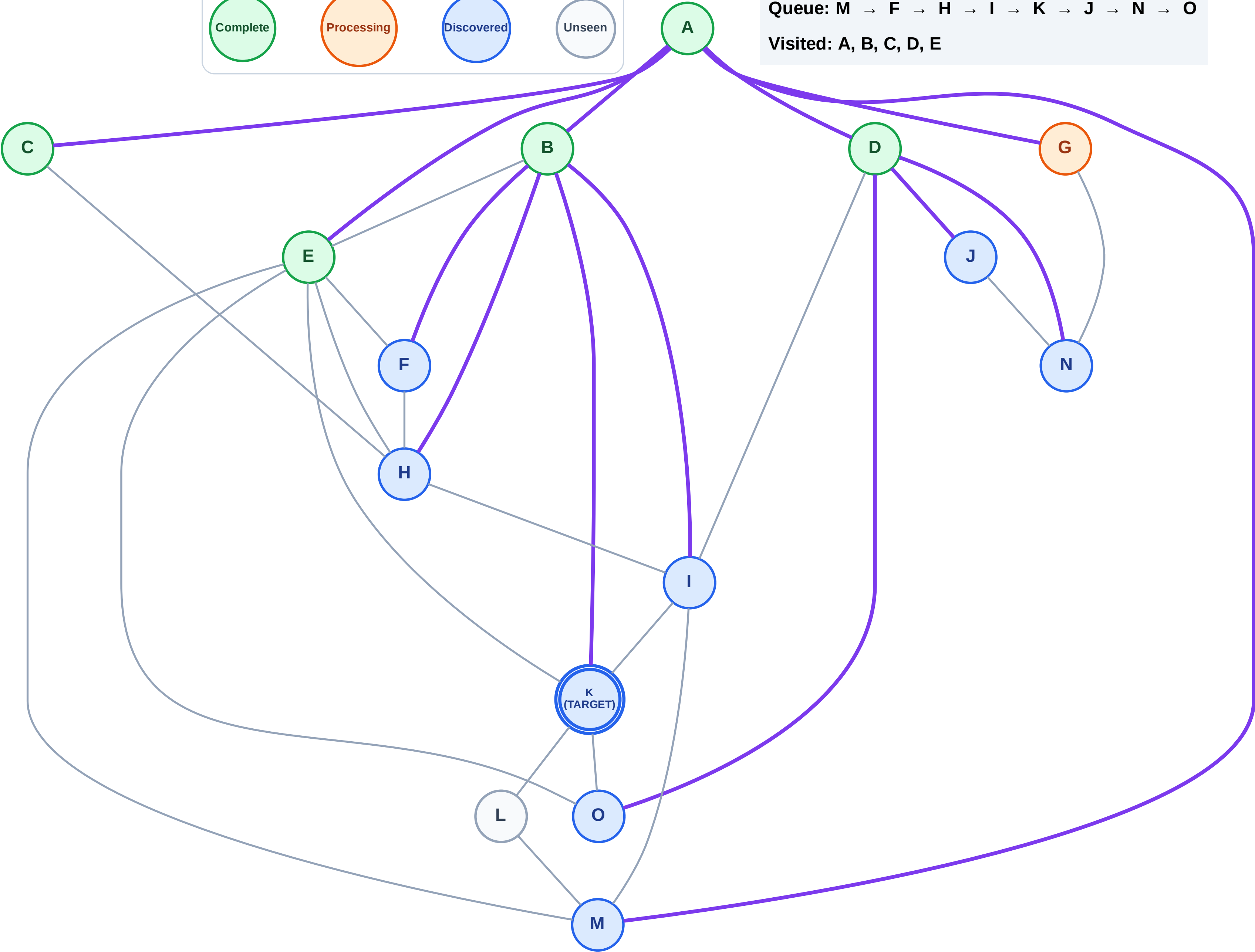
Processing

Discovered

Unseen

Queue: M → F → H → I → K → J → N → O

Visited: A, B, C, D, E



BFS Traversal

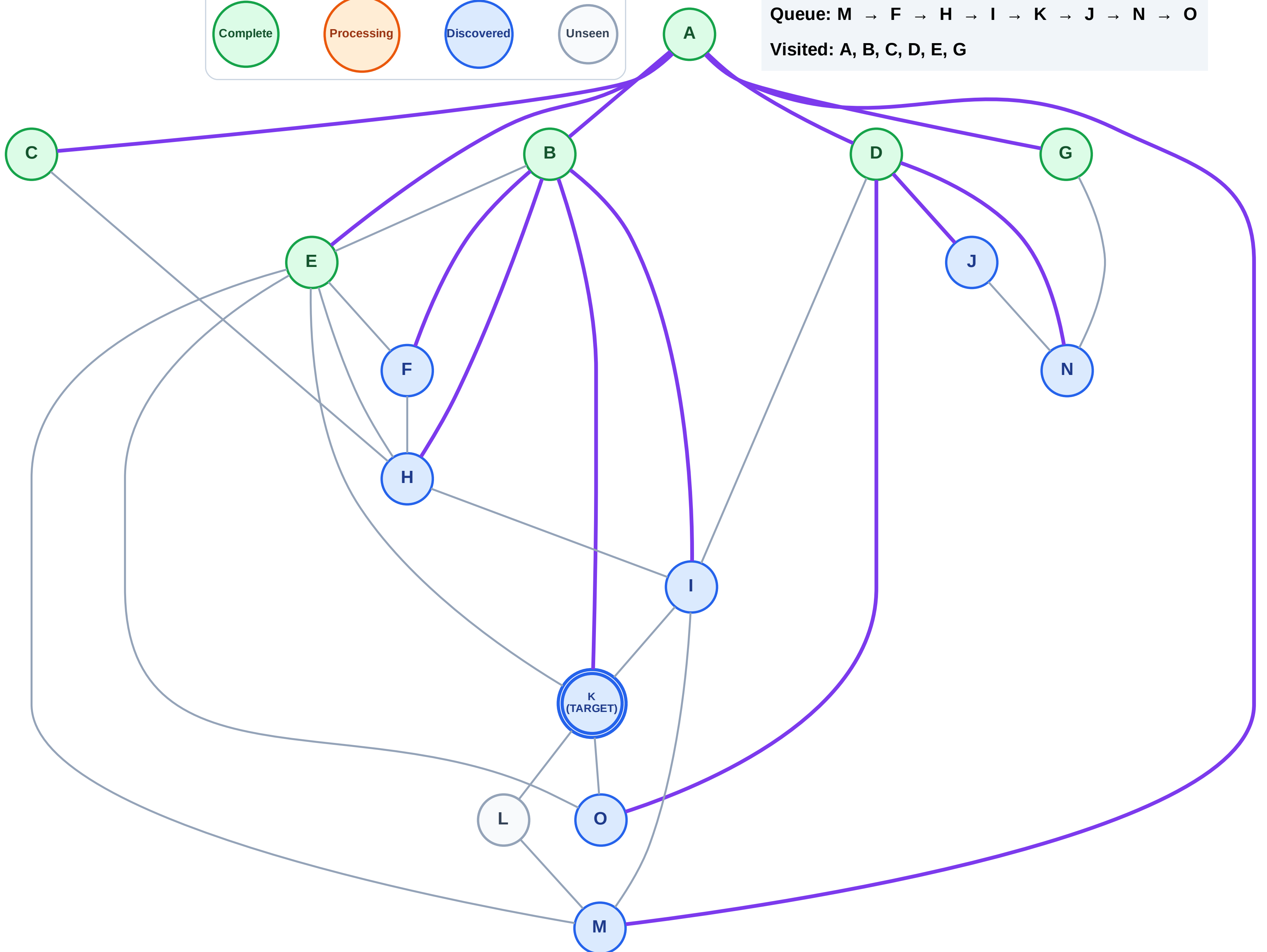
Step 13: No new neighbors from G

Legend



Queue: M → F → H → I → K → J → N → O

Visited: A, B, C, D, E, G



BFS Traversal

Step 14: Process node M

Legend

Complete

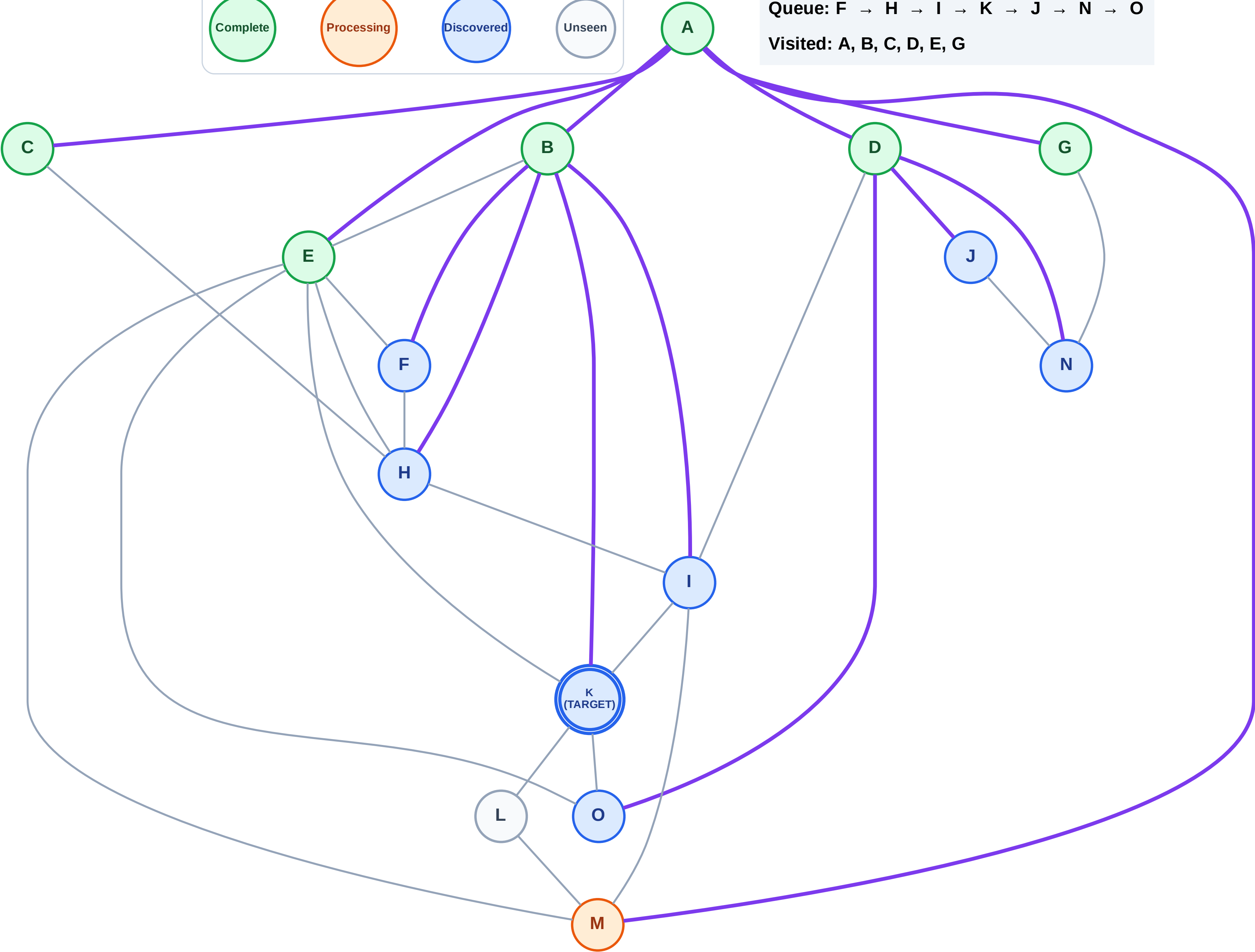
Processing

Discovered

Unseen

Queue: F → H → I → K → J → N → O

Visited: A, B, C, D, E, G



BFS Traversal

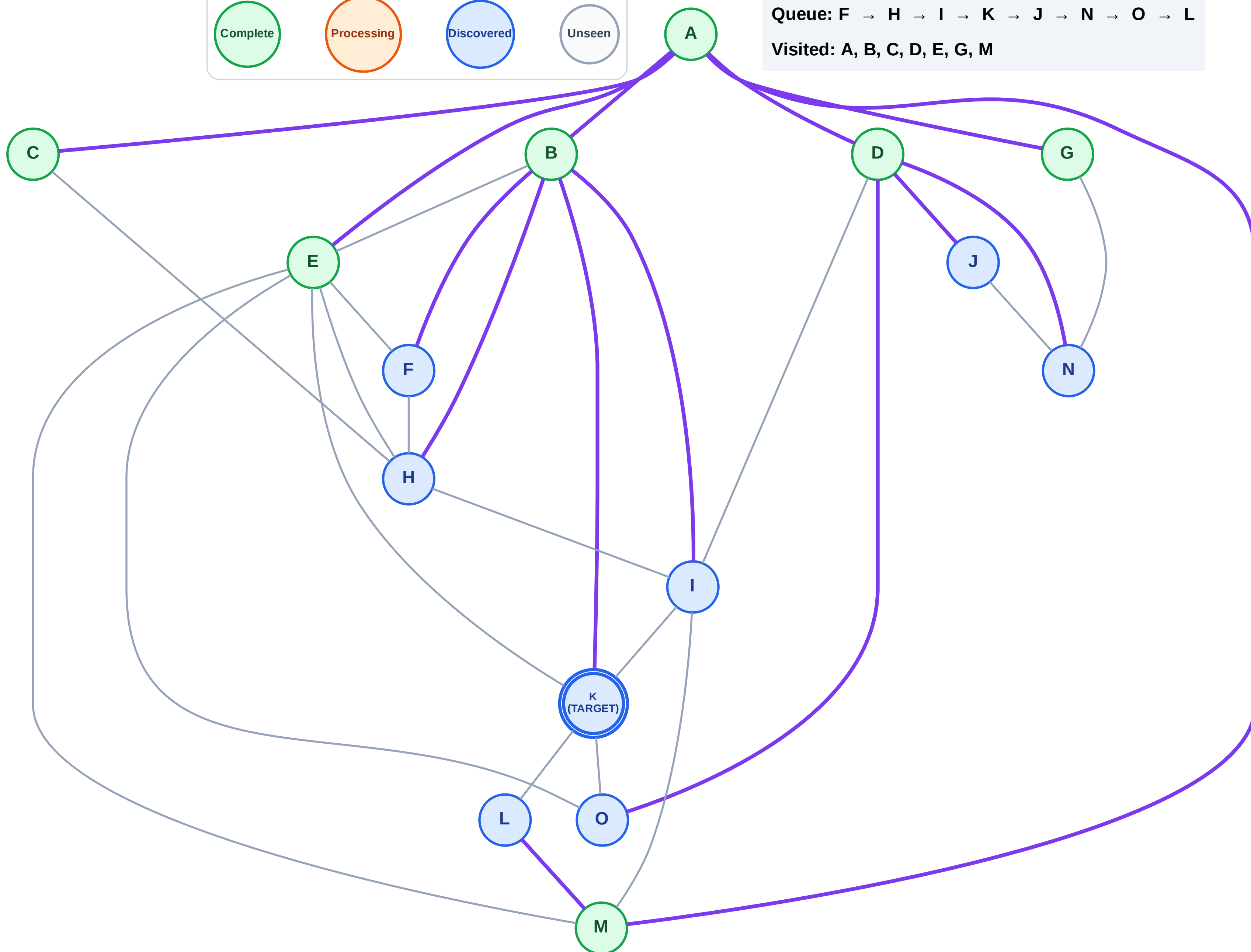
Step 15: Discover L from M

Legend



Queue: F → H → I → K → J → N → O → L

Visited: A, B, C, D, E, G, M



BFS Traversal

Step 16: Process node F

Legend

Complete

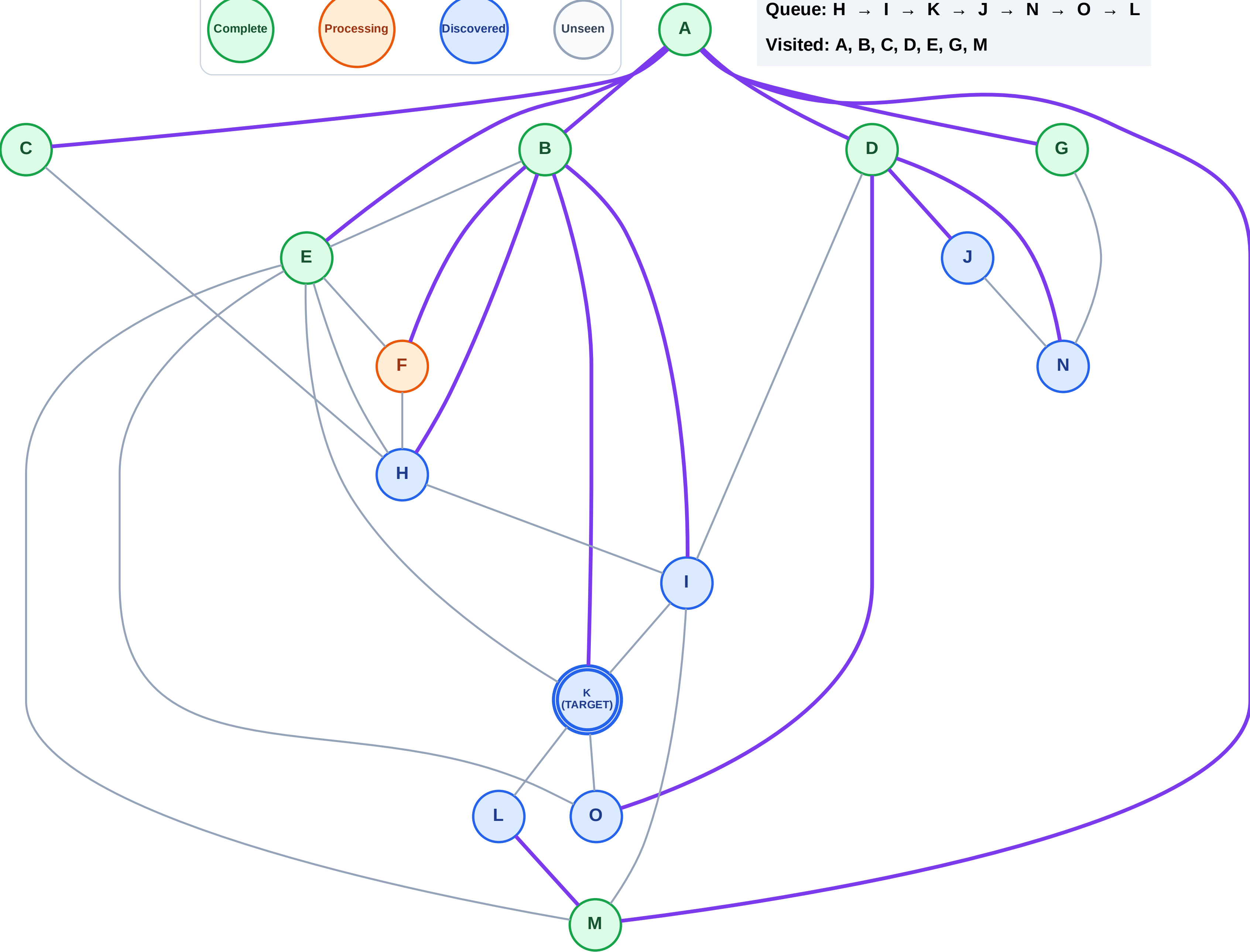
Processing

Discovered

Unseen

Queue: H → I → K → J → N → O → L

Visited: A, B, C, D, E, G, M



BFS Traversal

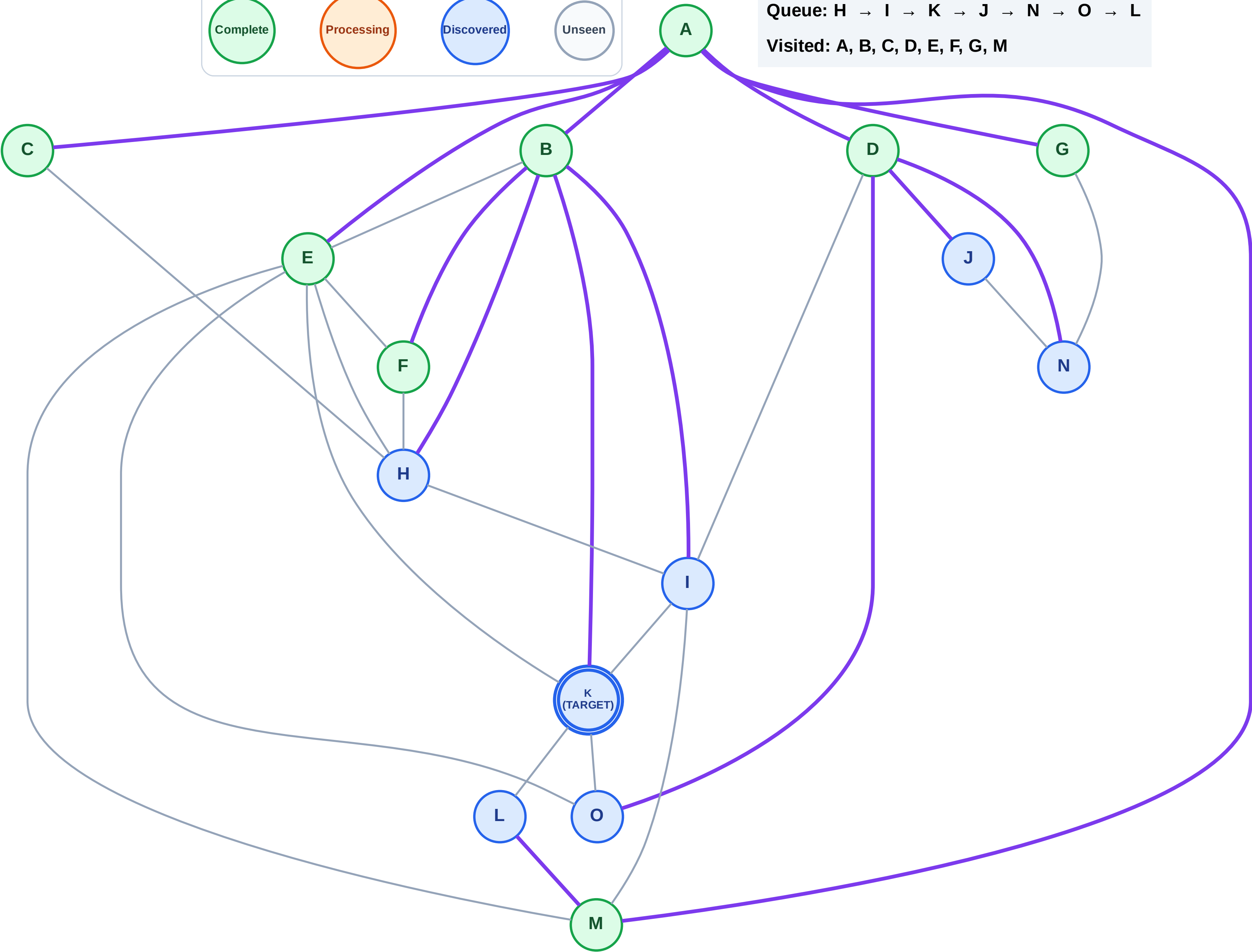
Step 17: No new neighbors from F

Legend



Queue: H → I → K → J → N → O → L

Visited: A, B, C, D, E, F, G, M



BFS Traversal

Step 18: Process node H

Legend

Complete

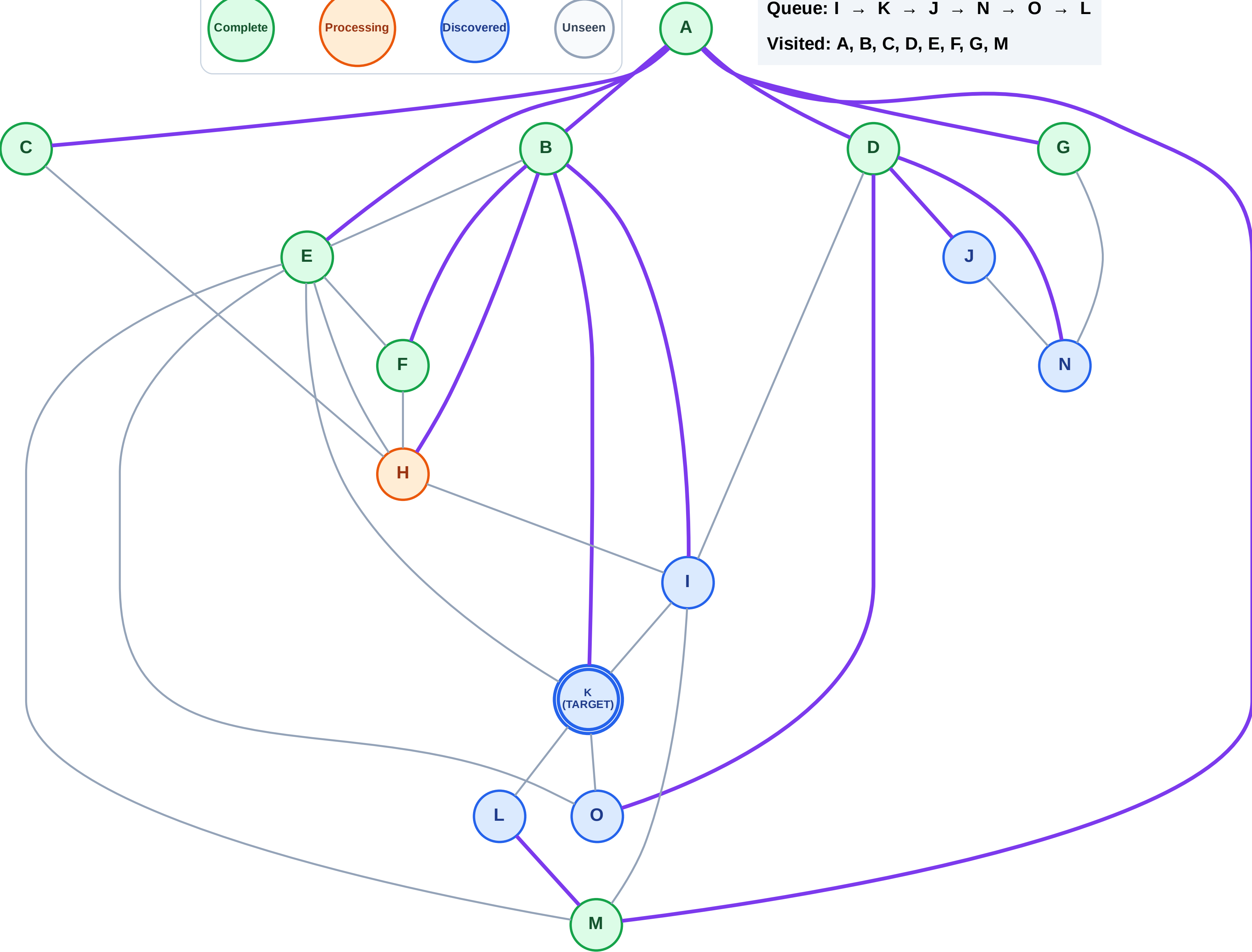
Processing

Discovered

Unseen

Queue: I → K → J → N → O → L

Visited: A, B, C, D, E, F, G, M



BFS Traversal

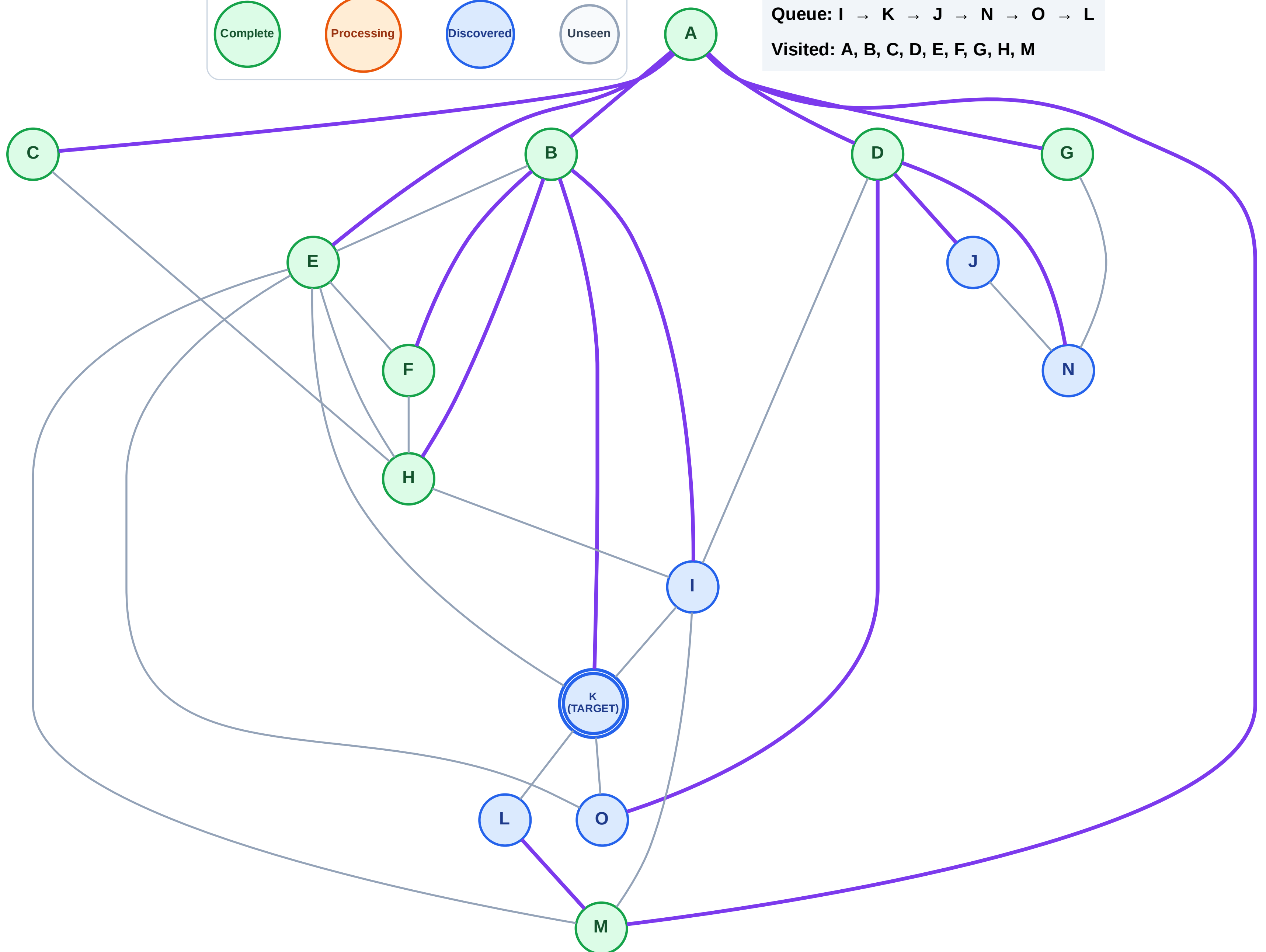
Step 19: No new neighbors from H

Legend



Queue: I → K → J → N → O → L

Visited: A, B, C, D, E, F, G, H, M



BFS Traversal

Step 20: Process node I

Legend

Complete

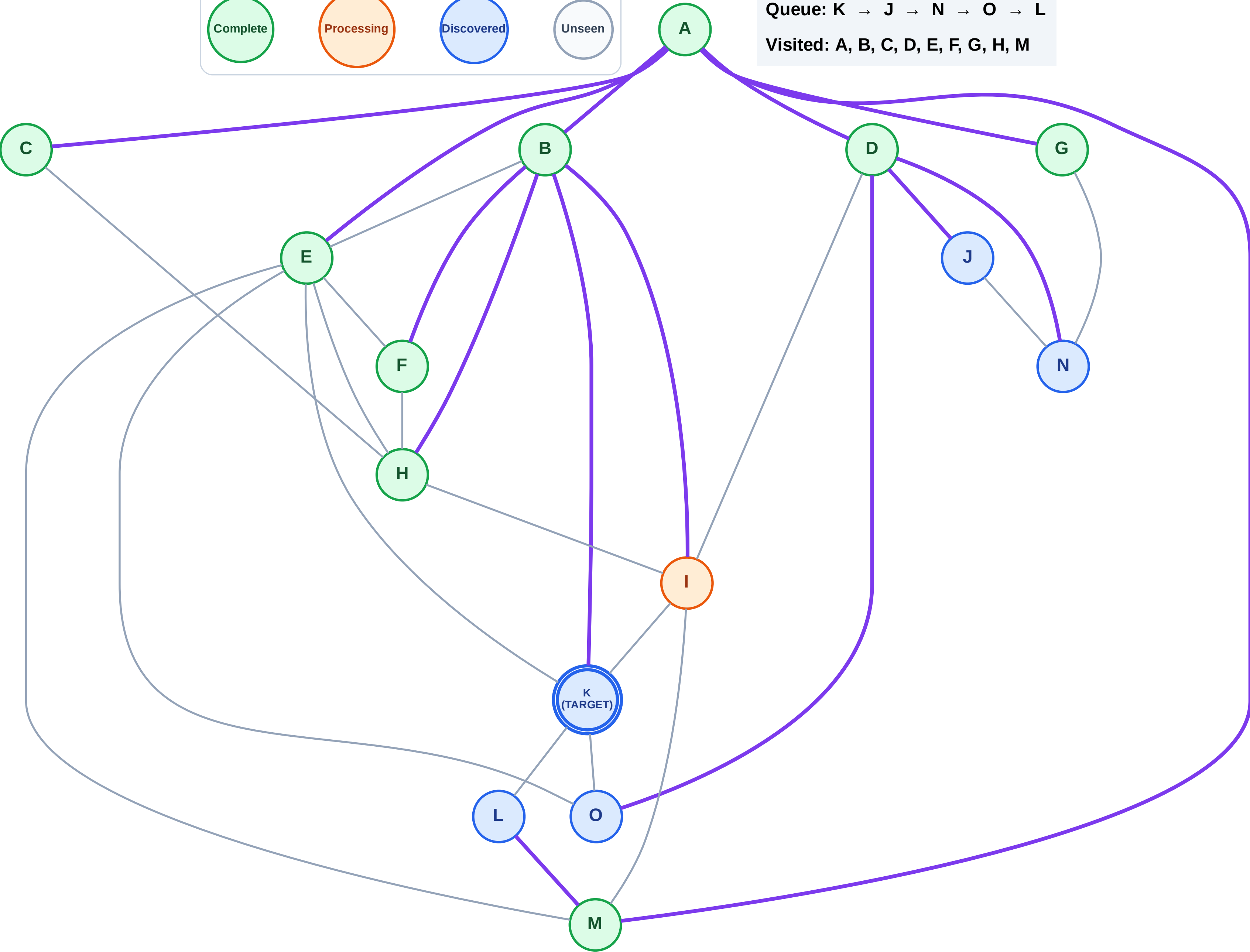
Processing

Discovered

Unseen

Queue: K → J → N → O → L

Visited: A, B, C, D, E, F, G, H, M



BFS Traversal

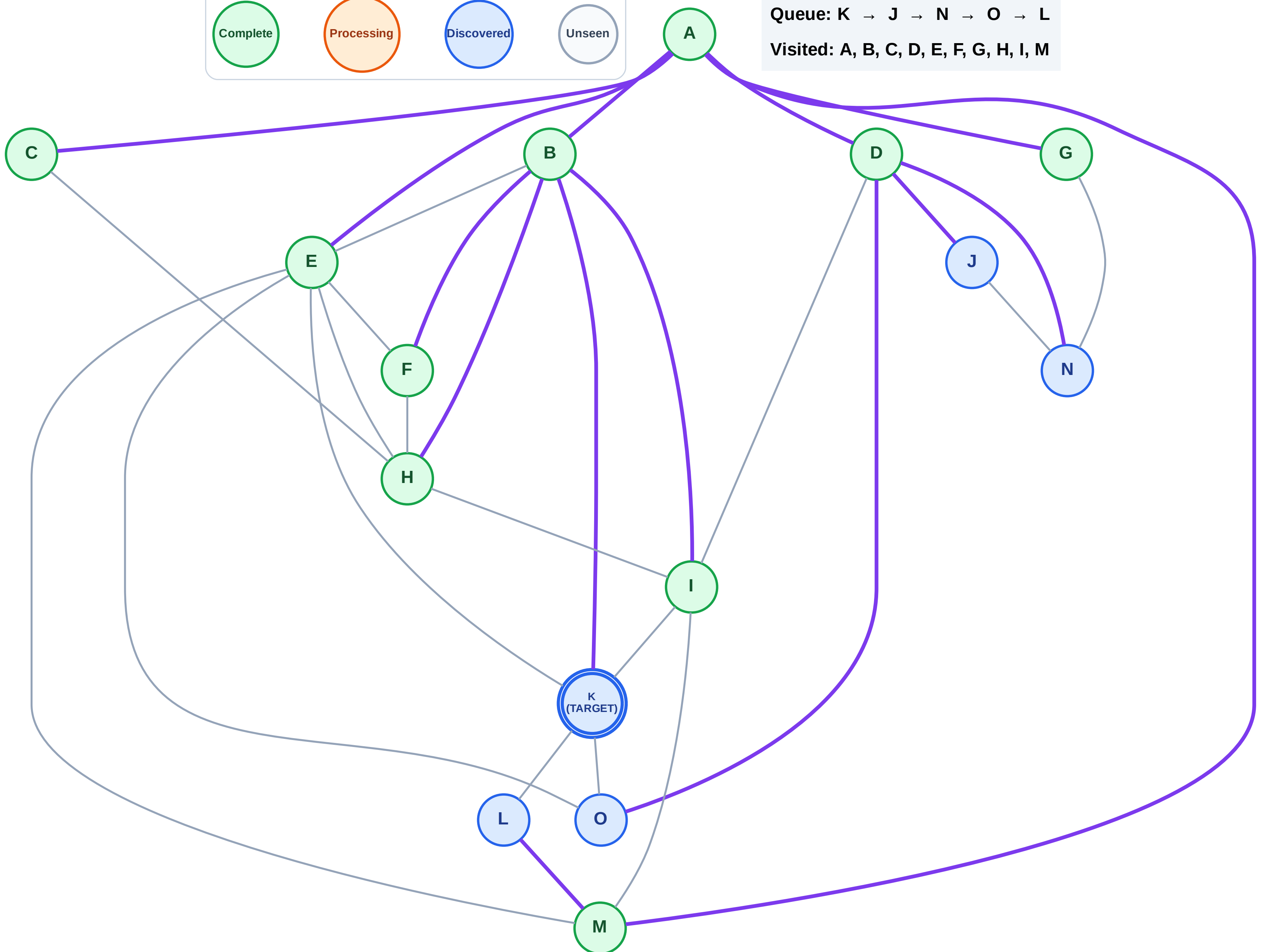
Step 21: No new neighbors from I

Legend



Queue: K → J → N → O → L

Visited: A, B, C, D, E, F, G, H, I, M



BFS Traversal

Step 22: Process node K

Legend

Complete

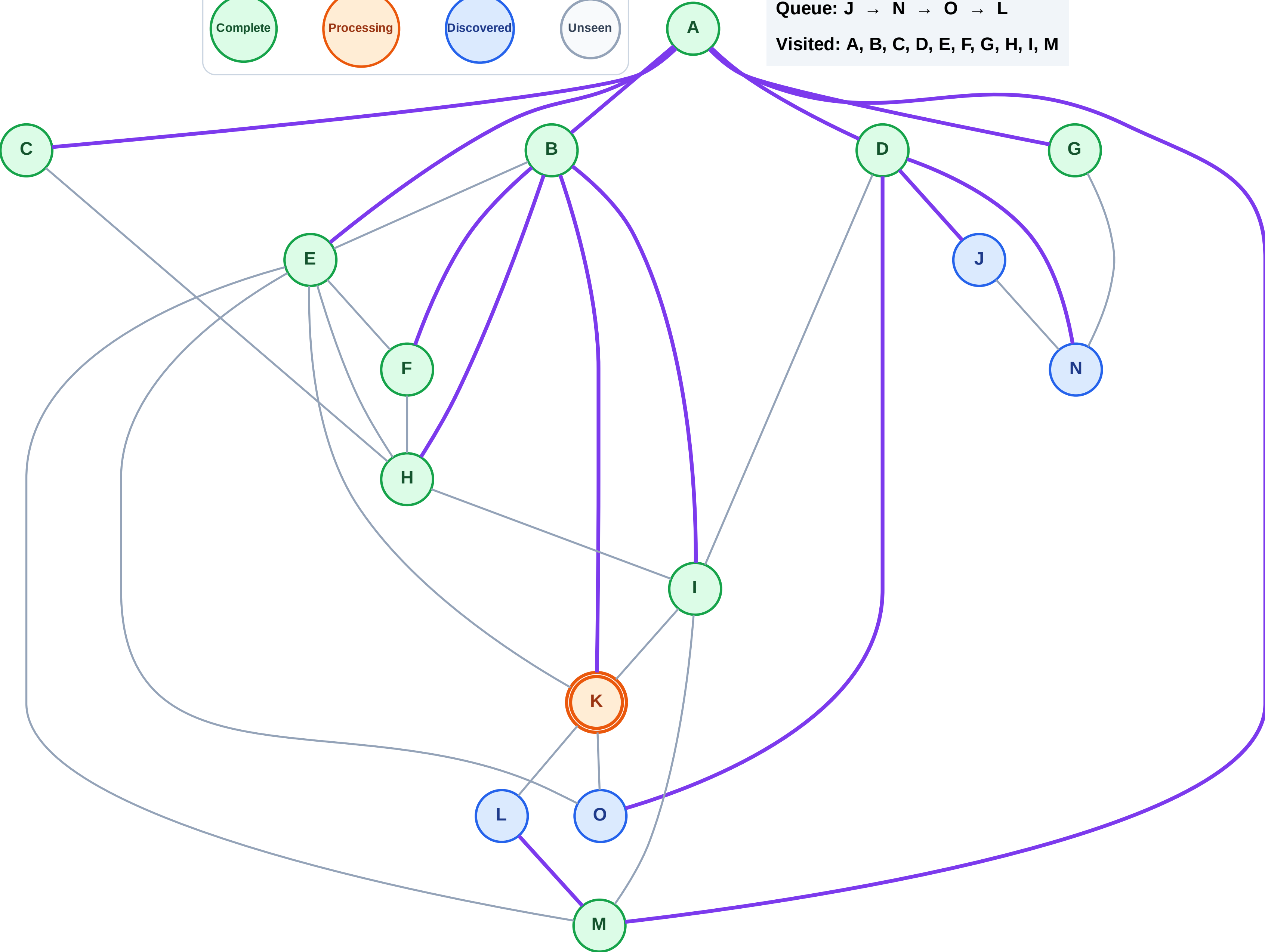
Processing

Discovered

Unseen

Queue: J → N → O → L

Visited: A, B, C, D, E, F, G, H, I, M



BFS Traversal

Step 23: TARGET NODE K REACHED! Path (2 hops): A → B → K

Legend

Complete

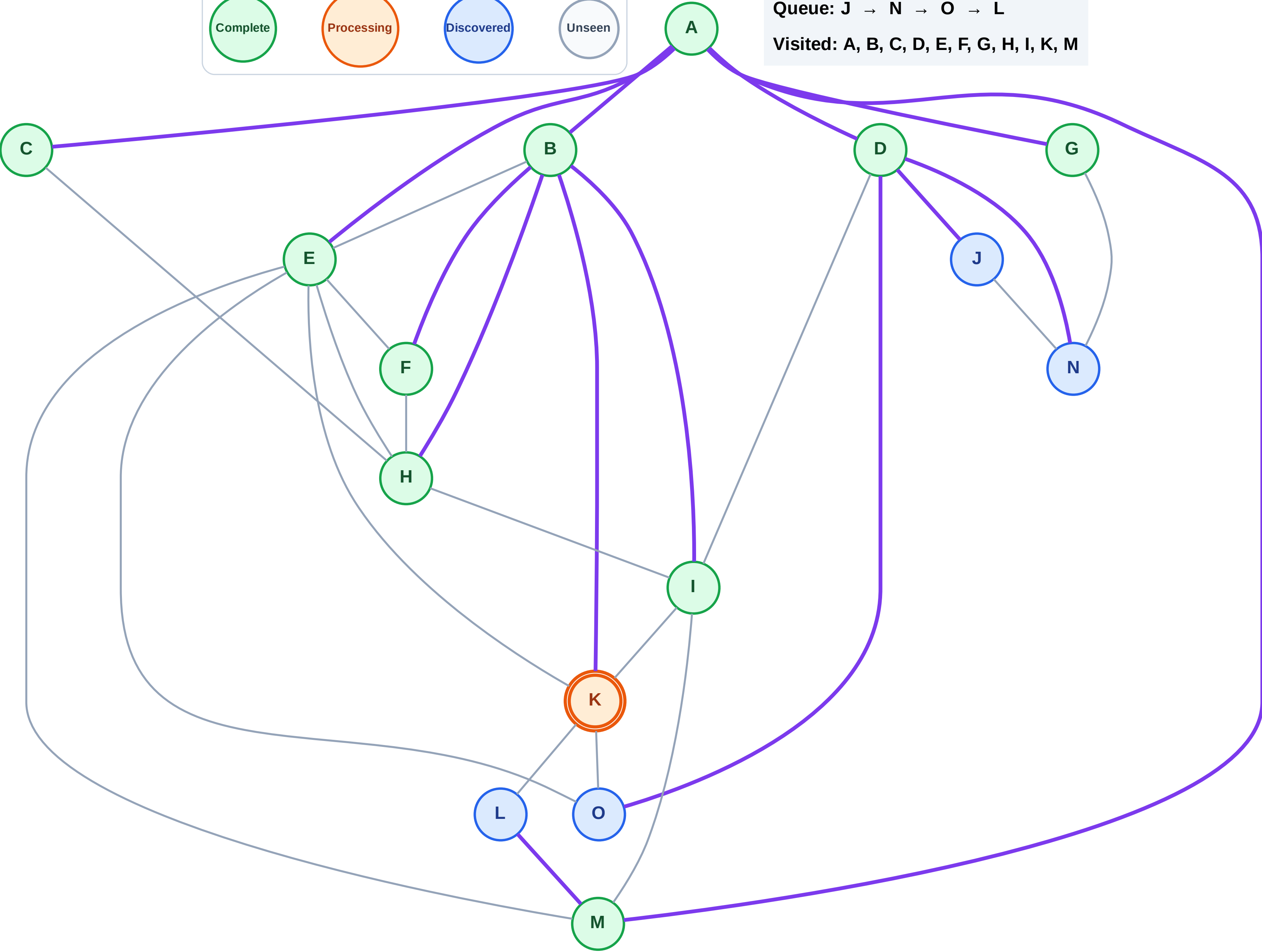
Processing

Discovered

Unseen

Queue: J → N → O → L

Visited: A, B, C, D, E, F, G, H, I, K, M



DFS Traversal

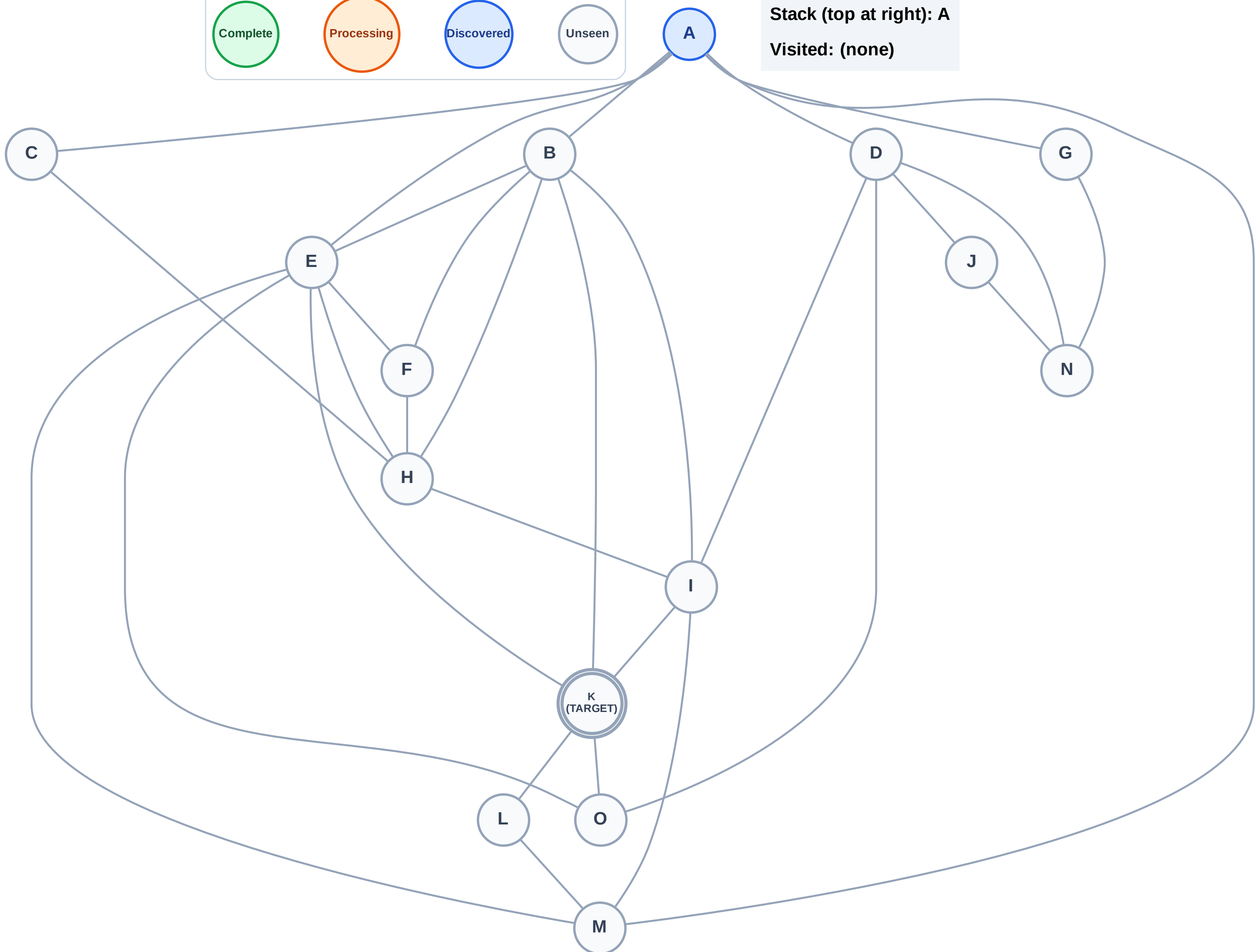
Step 1: Start at A (Seeking target node K)

Legend



Stack (top at right): A

Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

Complete

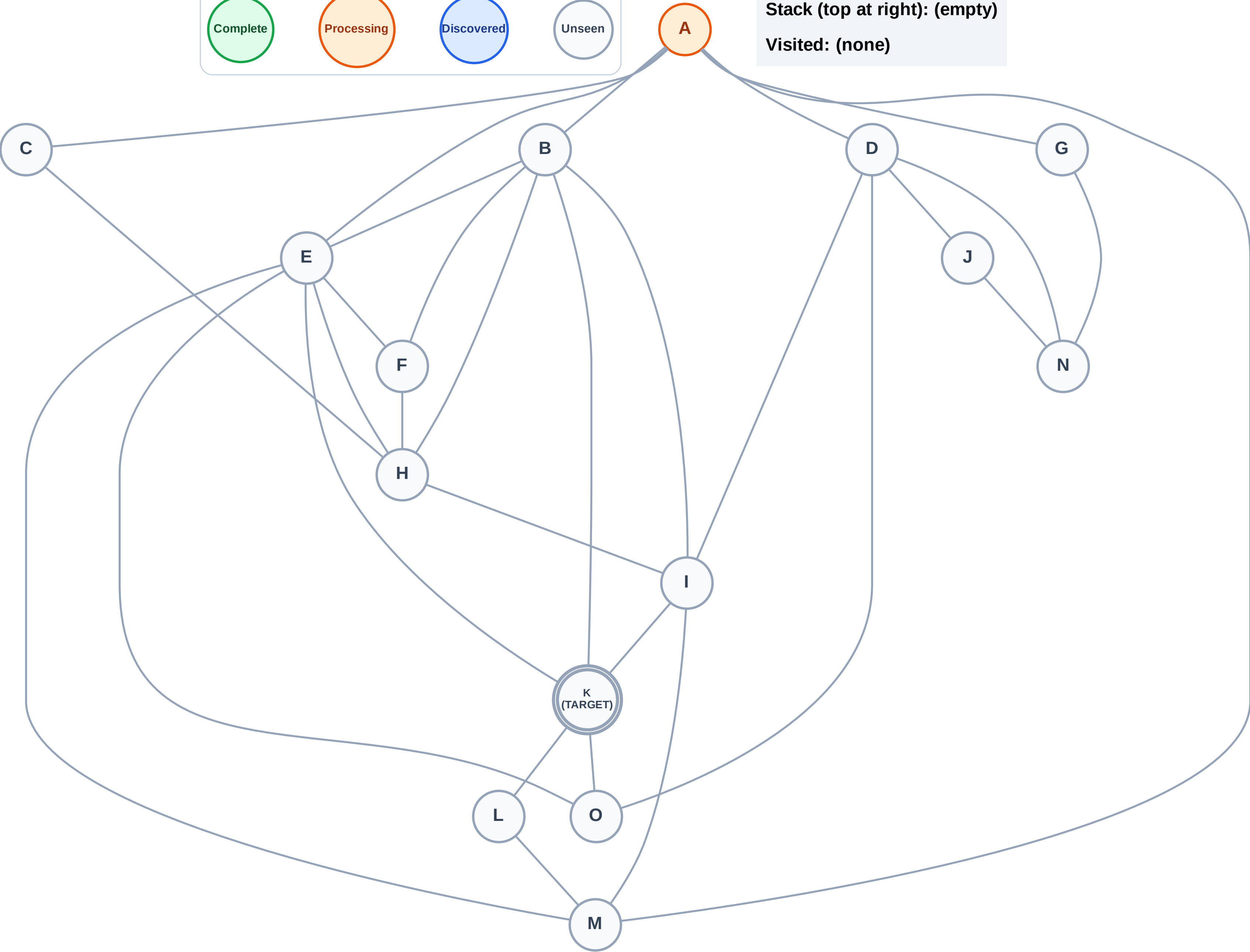
Processing

Discovered

Unseen

Stack (top at right): (empty)

Visited: (none)



DFS Traversal

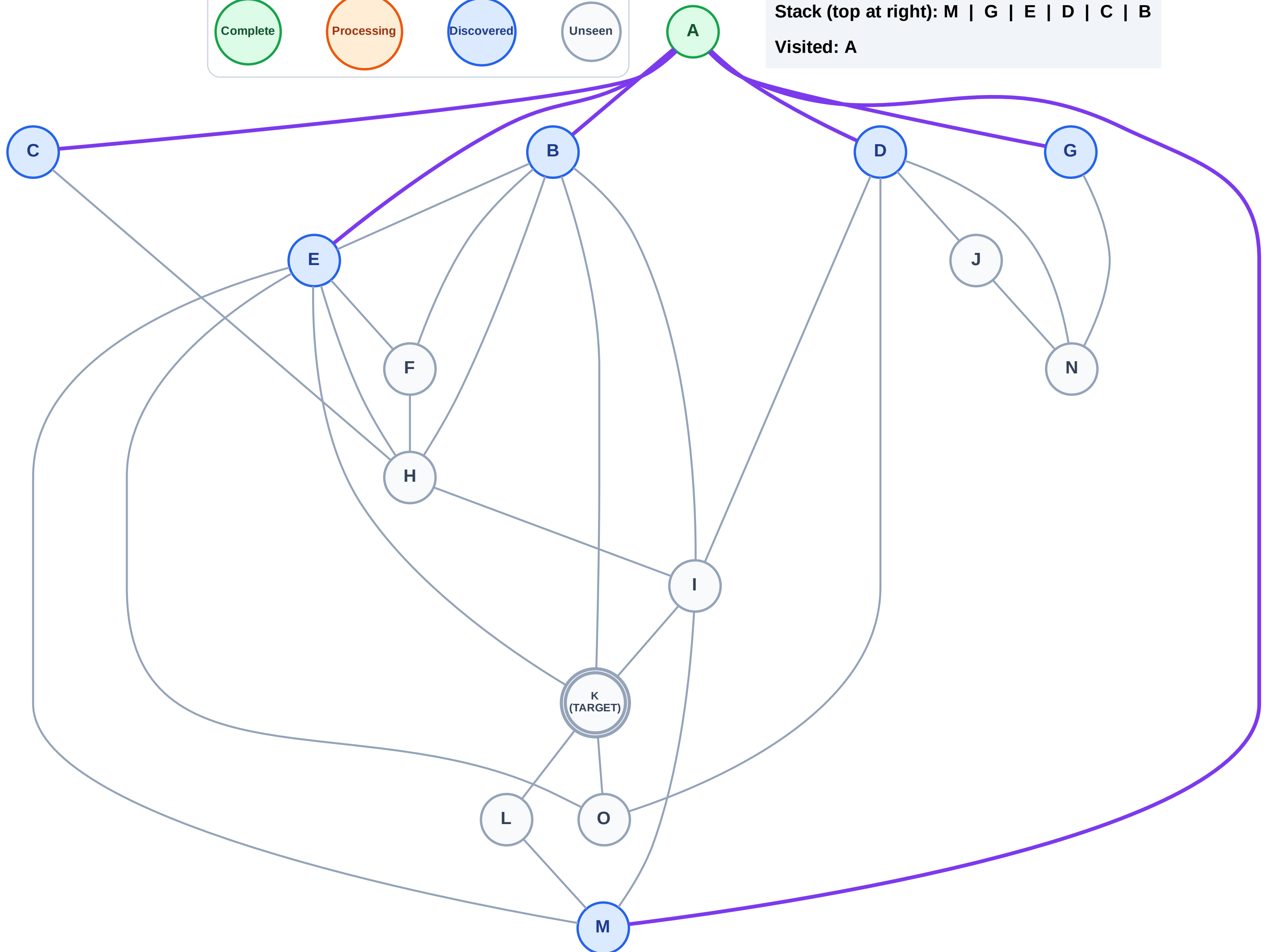
Step 3: Discover M, G, E, D, C, B from A

Legend



Stack (top at right): M | G | E | D | C | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

Complete

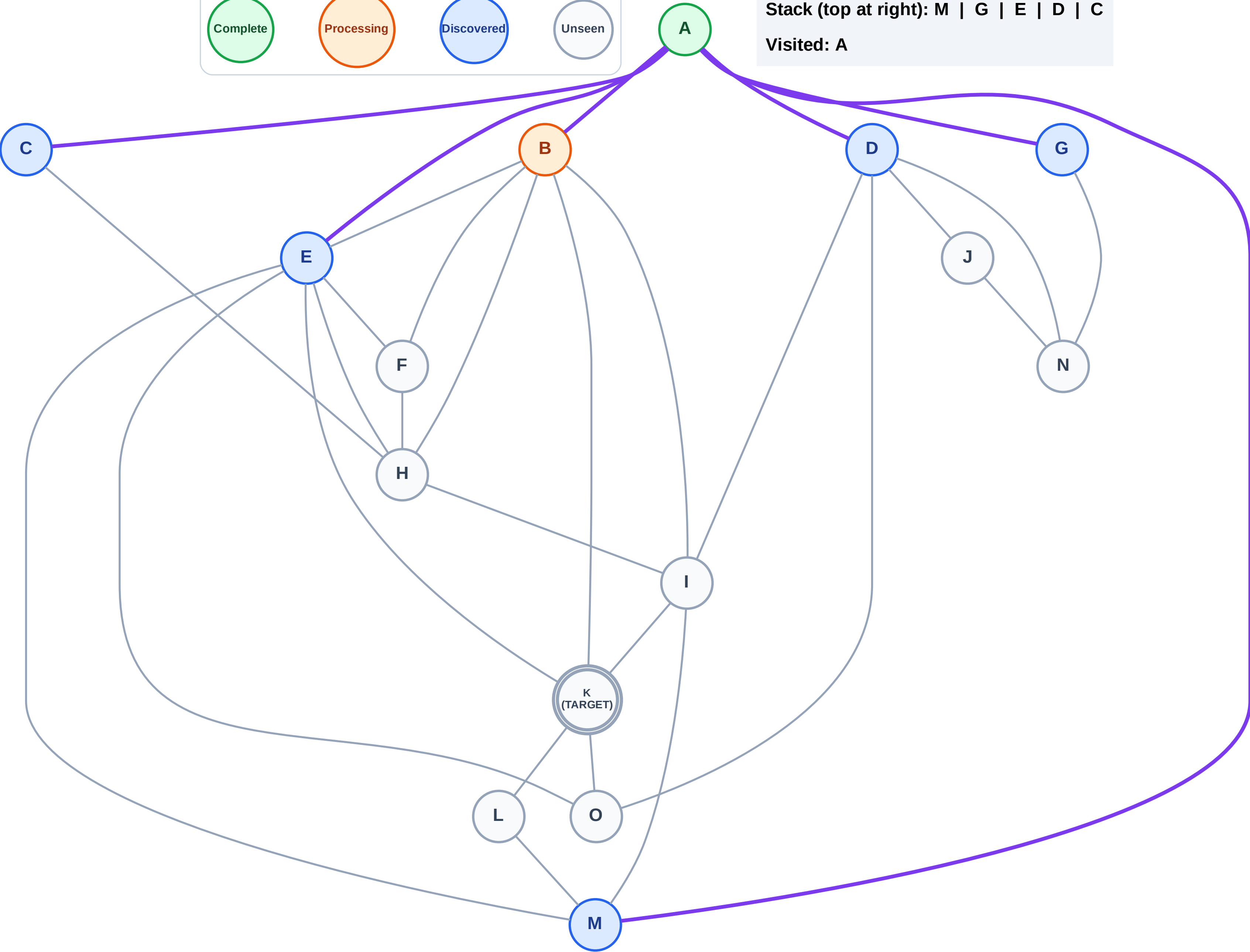
Processing

Discovered

Unseen

Stack (top at right): M | G | E | D | C

Visited: A



DFS Traversal

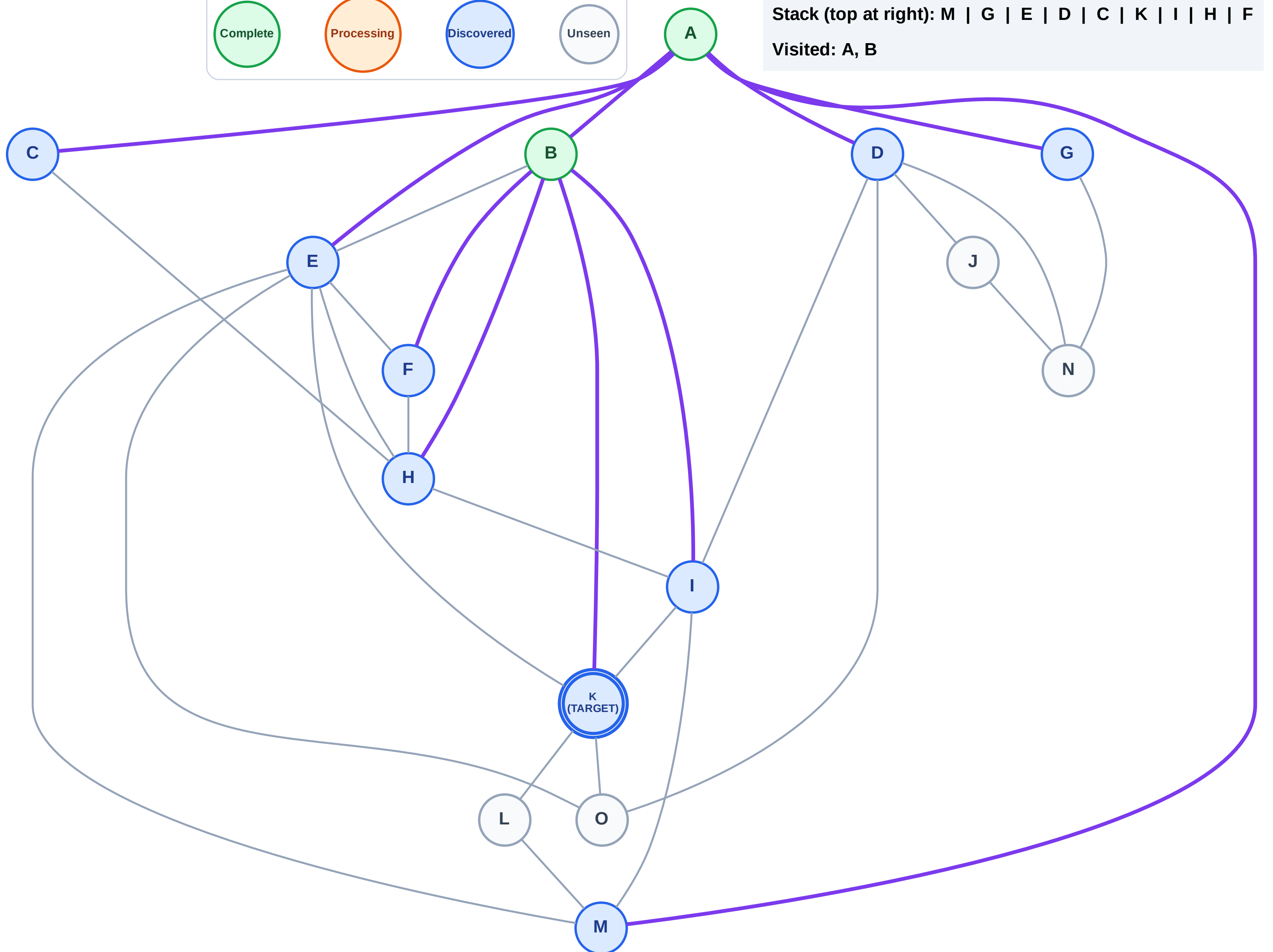
Step 5: Discover K, I, H, F from B

Legend



Stack (top at right): M | G | E | D | C | K | I | H | F

Visited: A, B



DFS Traversal

Step 6: Process node F

Legend

Complete

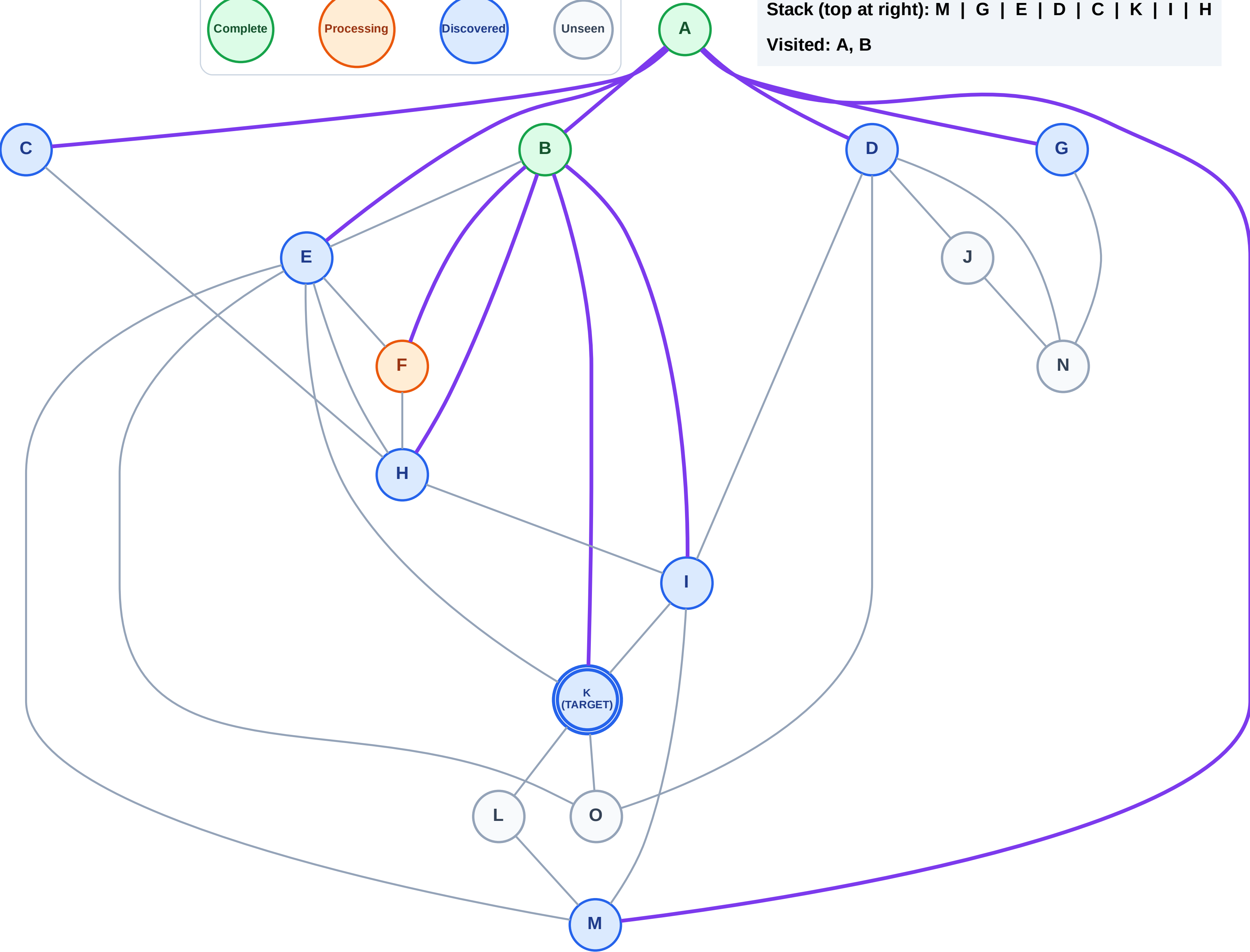
Processing

Discovered

Unseen

Stack (top at right): M | G | E | D | C | K | I | H

Visited: A, B



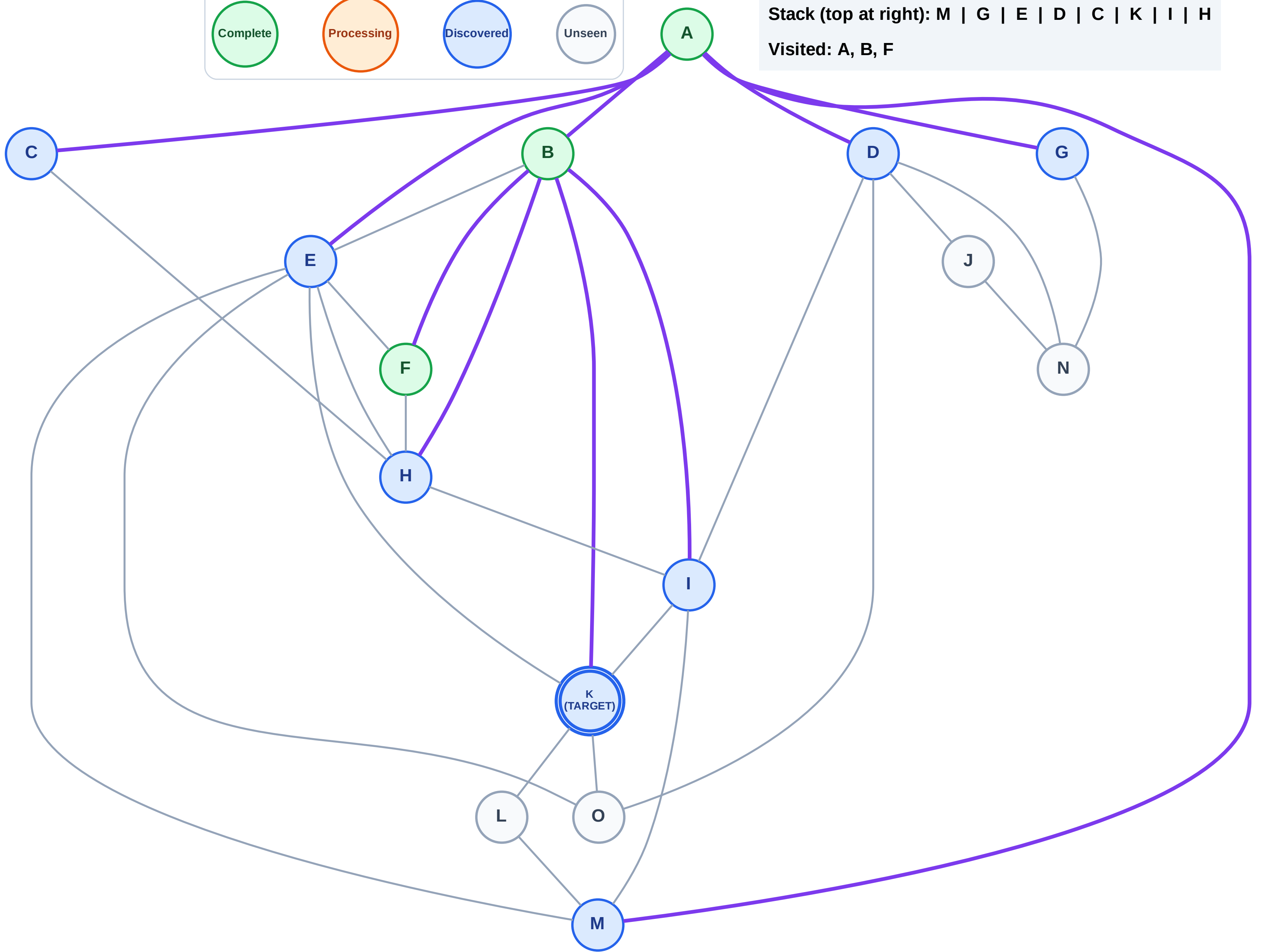
DFS Traversal

Step 7: No new neighbors from F

Legend



Stack (top at right): M | G | E | D | C | K | I | H
Visited: A, B, F



DFS Traversal

Step 8: Process node H

Legend

Complete

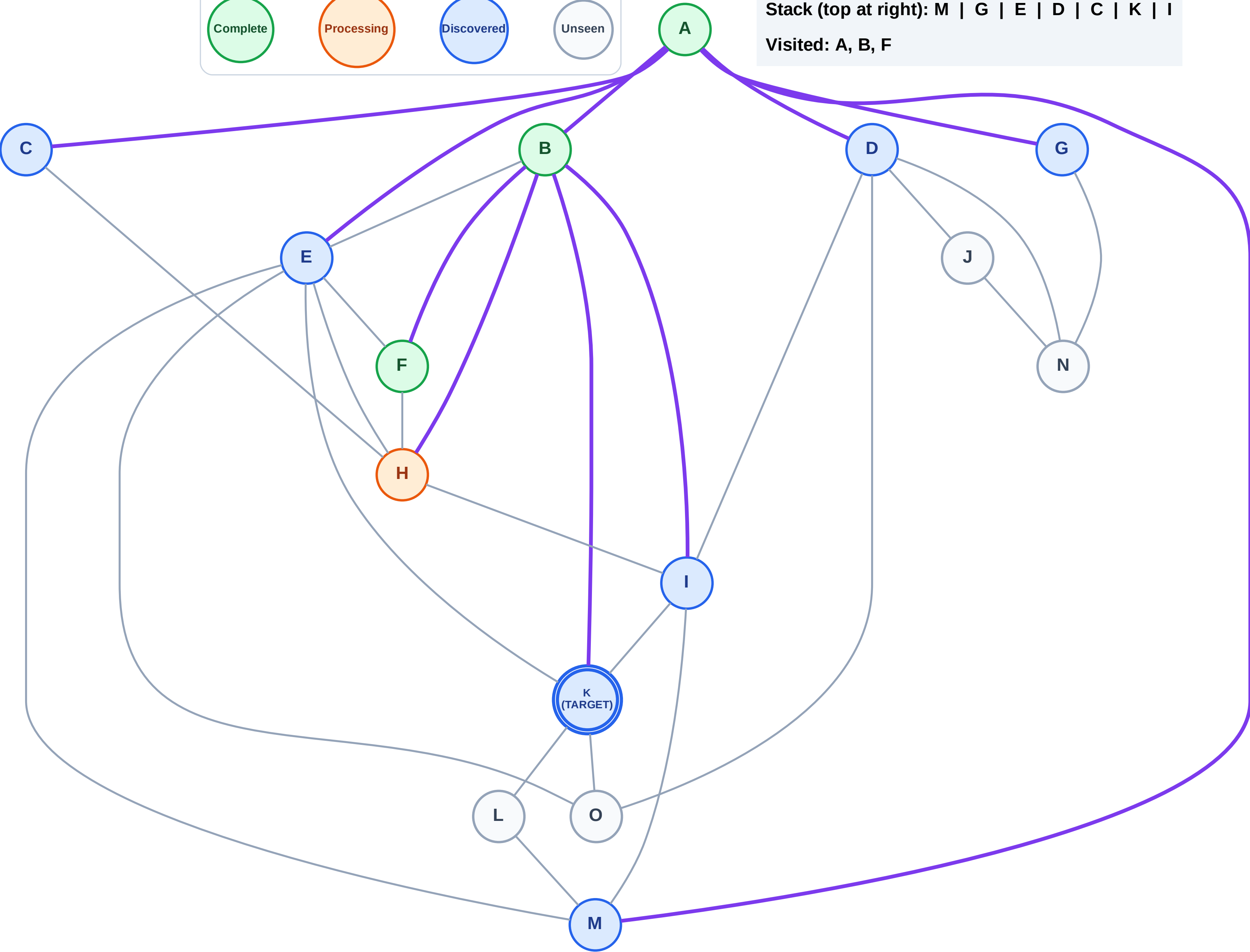
Processing

Discovered

Unseen

Stack (top at right): M | G | E | D | C | K | I

Visited: A, B, F



DFS Traversal

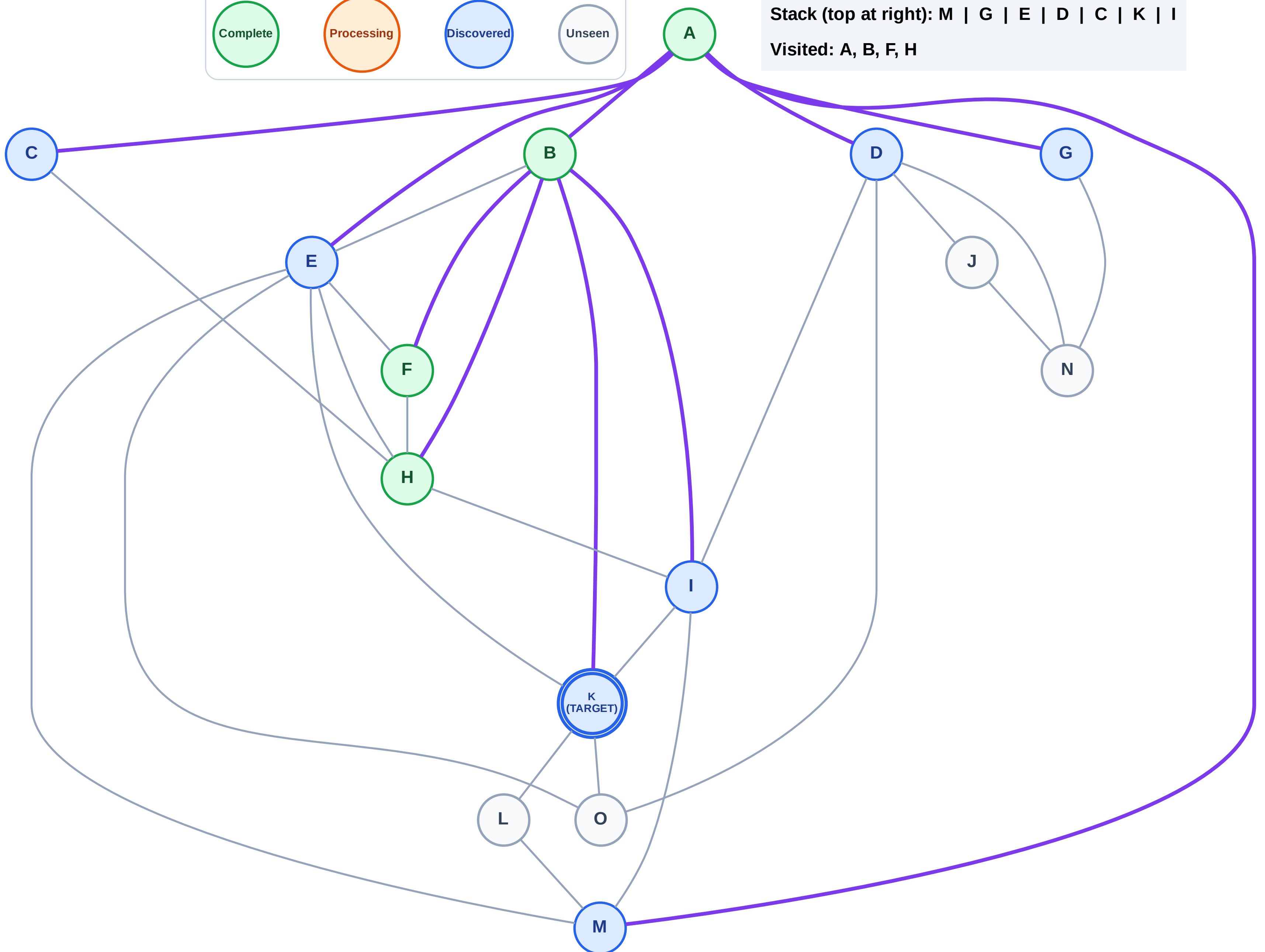
Step 9: No new neighbors from H

Legend



Stack (top at right): M | G | E | D | C | K | I

Visited: A, B, F, H



DFS Traversal

Step 10: Process node I

Legend

Complete

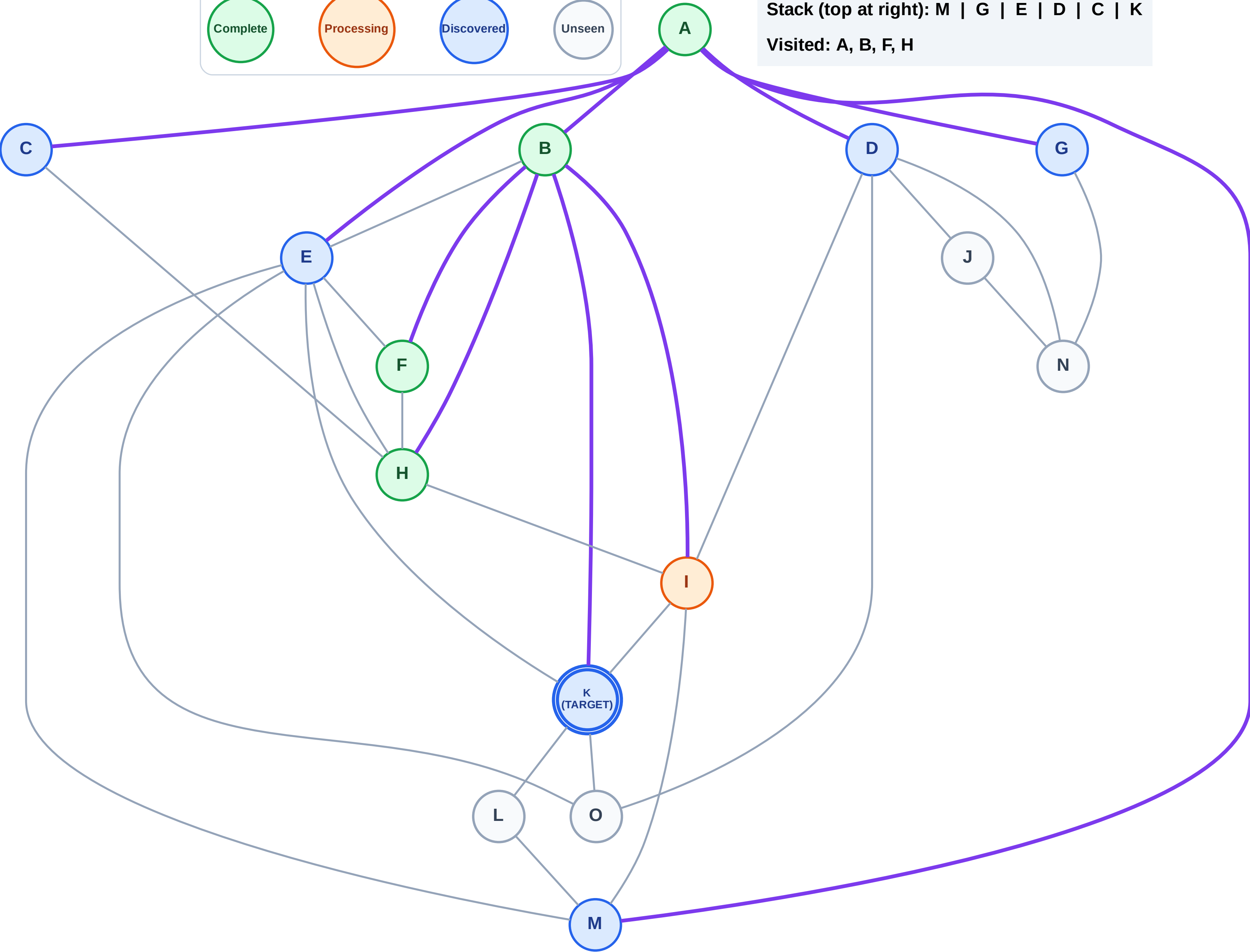
Processing

Discovered

Unseen

Stack (top at right): M | G | E | D | C | K

Visited: A, B, F, H



DFS Traversal

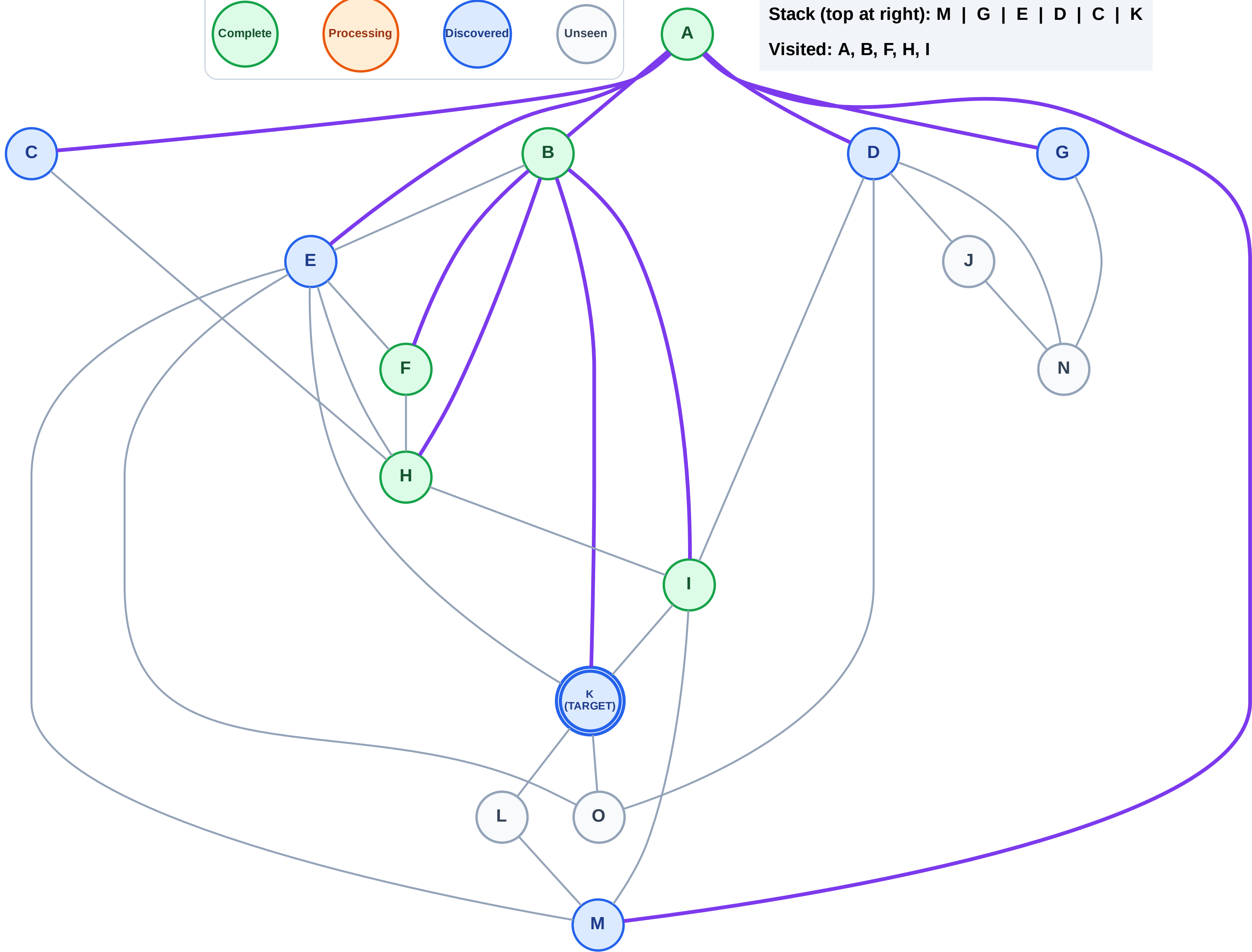
Step 11: No new neighbors from I

Legend



Stack (top at right): M | G | E | D | C | K

Visited: A, B, F, H, I



DFS Traversal

Step 12: Process node K

Legend

Complete

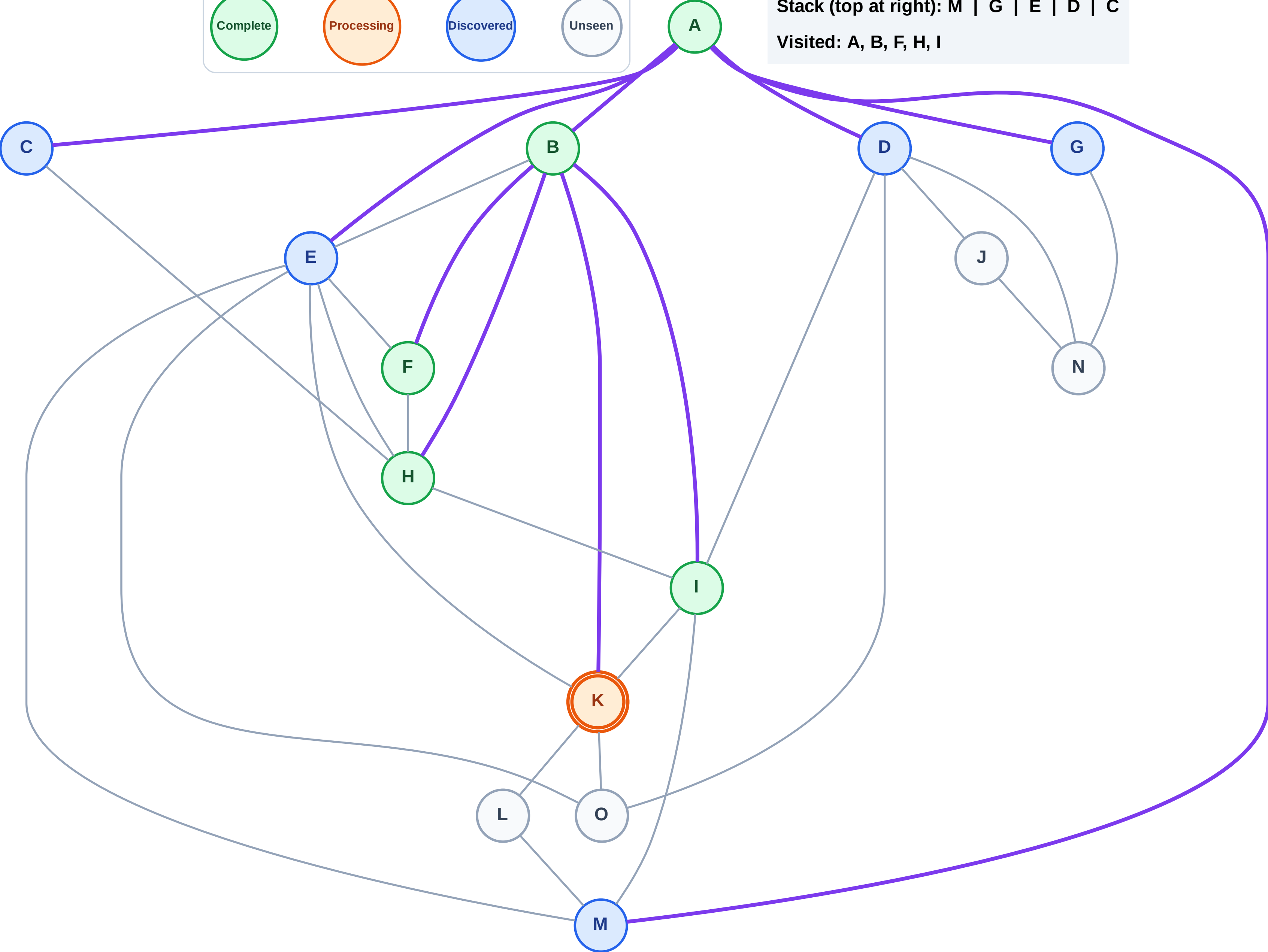
Processing

Discovered

Unseen

Stack (top at right): M | G | E | D | C

Visited: A, B, F, H, I



DFS Traversal

Step 13: TARGET NODE K REACHED! Path (2 hops): A → B → K

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): M | G | E | D | C

Visited: A, B, F, H, I, K

